GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We study the exact classical memory required to reproduce the laws of a causal experiment. For compactly reversible finite-dimensional responses, every finite stochastic realization reduces to a permutation core and at most one counted initializer. We prove that, for every finite observable group, sufficiently small positive terminal contrast makes the least core equal to the least dimension of a permutation module containing the observable module. All memory thresholds along the contrast line are attained. For groups whose subgroups are all normal, an absorption theorem reduces the exact optimization to quotient linear programs with explicit payoff duals and rational thresholds for rational data. Sharp examples show positivity transitions invisible to the group and observable module. The retained theory supplies a common exact classical simulator for general finite quantum instruments under a uniform interior condition on every posterior. We strengthen the distinction between this condition and average mixing: a stationary qubit family has an exactly depolarizing unconditioned channel, one-dependent observations and Hankel rank three, but its minimum memory is exactly q with mixed initialization and $q+1$ with a counted pure start when its conditional phase has finite order $q \geq 3$; a non-root-of-unity phase requires infinite memory. These results connect predictive compression, positive realization and finite-resource causal comparison. Complete stopping, variational, dual, reset, rank-constrained discrimination, spectral and geometric results retain their separate hypotheses and proofs.

1. Introduction

A predictive state records everything about a past experiment that can affect its future laws. Its linear dimension need not measure the memory required to realize those laws with nonnegative transition probabilities. The distinction matters in causal experiments: commands can change what is observed next, observation changes the state on which later commands act, and the resources used to retain or communicate information must themselves be counted. This paper studies that relation between predictive compression and exact finite realization, within a broader theory of causal resource comparison and optimal discrimination.

Two forms of observation will be used. A terminal response assigns a probability row $b(w)$ to every finite command word w. An instrument response assigns a joint probability to every finite command–outcome history. A classical realization has one fixed collection of stochastic tables and a finite set of live labels. Unless a mixed-start comparison is explicitly stated, the initial label is pure and included in the count. The tables may use arbitrary real probabilities; a lower bound therefore cannot be evaded by increasing numerical precision. Their description and execution costs are separate resources.

Reversible responses and the positivity gap. Let V_b be the span of residual functions $b^w(z) = b(wz)$, with dimension d. Suppose that the command shifts are invertible on V_b and generate a group with compact closure G_b. This compact reversibility condition holds for prepared finite-dimensional unitary command families after all directions invisible to future tests have been removed. The finite

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realization criterion, Theorem 3.1, says that an exact finite classical realization exists precisely when G_b is finite.

Theorem 4.1 gives the resource structure of every such realization. It replaces continuing stochastic dynamics by a finite permutation action and at most one transient initializer, without increasing the number of live labels. The original transition matrices need not form a group. The proof compresses their observable action through recurrent classes of words which act as the identity on V_b. If N is the number of deterministic residuals and $\kappa_G(b)$ the minimum size of a probability factorization $b(g) = \lambda P_g D$, then

$$m(b) = \min\{N, 1 + \kappa_G(b)\}.$$

The bounded decoder and the initial mixture are essential. The least size $\nu_G(V_b)$ of a permutation module containing V_b is a lower bound for $\kappa_G(b)$, and can be strictly smaller.

We determine how this positivity gap changes under terminal readout contrast. Let π be the group-averaged probability row and put $b_t = \pi + t(b - \pi)$, $0 \le t \le 1$. For every finite group and every finite outcome set, Theorem 5.1 proves

$$\kappa_G(b_t) = \nu_G(V_b), \qquad m(b_t) = \min\{N, 1 + \nu_G(V_b)\}$$

for all sufficiently small positive t. Throughout $t > 0$, the group, module and residual cardinality stay unchanged. The proof constructs a strictly positive preparation in a smallest containing permutation module and controls the decoder explicitly. Theorem 5.2 then gives a complete finite filtration by attained thresholds. Its general algebraic computability is distinct from a model-specific closed form.

For every group whose subgroups are normal, including all finite abelian groups, Theorem 6.2 makes the optimization linear. Within-orbit preparation can be absorbed into a quotient decoder; only a mixture among distinct quotient orbits remains. Theorem 6.3 gives necessary and sufficient payoff inequalities, strict exclusion certificates and exact contrast endpoints. Rational data have rational optimal controllers and rational thresholds. For abelian groups, character support determines the small-contrast module bound. At higher contrast, positivity adds a genuine obstruction: a four-command binary family has $d = 6$ and sixteen residuals at every $t > 0$, but its minimum is eleven labels for $t \le 3/5$ and thirteen for $t > 3/5$. Explicit payoff witnesses also give a robust neighbourhood of the larger-memory regime. The retained prepared-qubit S_3 model has an exact four-to-six transition at $2/3$.

Observed instruments and conditional memory. Reversibility is only one way in which response memory is constrained. The retained Theorem S8.2 treats known finite quantum instruments on $\mathbb{C}^h$ whose every positive normalized branch state is at least $\eta I/h$. For one fixed known prepared state it constructs a simulator with at most

$$1 + \left\lceil (4h/\eta)^{h-1} \right\rceil$$

live labels. This theorem was proved before the present revision, together with sparse branch rows, several-step interiority and a state-positive noise exponent. A finite physical dictionary contains the whole posterior region; branchwise barycentric expansions intertwine each completely positive map with a nonnegative outcome matrix. Consequently the same tables reproduce all adaptive visible histories, terminal POVMs, almost surely finite stopped laws and nonnegative history costs. No state-independent outcome probabilities or commuting-command assumption is required. Applying depolarizing noise after each observed branch supplies the hypothesis for arbitrary finite instruments. A complementary common-contraction theorem gives an explicit finite positive realization of supplied linear responses under one norm certificate for all commands.

Average mixing is insufficient for the instrument conclusion. Theorem 9.1 constructs a stationary qubit family with an exactly depolarizing unconditioned channel, uniformly positive outcome probabilities, one-dependent observations and Hankel rank three. If its conditional phase has order $q \ge 3$, its exact classical memory is q with a free mixed start and $q + 1$ with a counted pure start.

A non-root-of-unity phase requires infinite memory. The upper uses one invariant polygon for all outcomes; the lower uses an observable cylinder whose relative peripheral spectrum forces a q-cycle. Equality in the cycle bound is rigid enough to rule out a pure q-label initialization. Thus neither linear rank, unconditioned mixing time nor finite dependence determines the positive realization resource. The exact finite and infinite cases lie in one fixed physical dimension and one fixed outcome alphabet.

Classical ingredients and contribution. Finite residual equivalence, stochastic automata, positive realization and observable operator models are classical [44, 48, 49, 45, 46, 47]. Schwarz, Flor and Brown developed the structure of nonnegative and stochastic matrix groups [61, 62, 60, 66]. Symmetric cone factorizations and extended formulations [67, 68], nonnegative-rank perturbations [69], and restricted nonnegative factorization [70] precede the present work. Their general mechanisms are credited. The assertions here concern the exact observable normal form with its initial charge, universal attainment of the permutation-module bound at small contrast, the normal-quotient linear reduction, and the resulting exact resource transitions.

Invariant-polytope simulation of quantum instruments and adaptive composition are explicitly developed by Okay [71]; pure-state nets and convex approximation of density matrices are established tools [72]. The earlier instrument results of this paper supply the posterior condition, quantitative physical dictionary and exact resource accounts, with complete proofs retained in Section S8. They are used here as established premises, not counted again as new results. Finite-dimensional quantum processes without finite classical realizations already occur in positive-realization theory [50]. Holroyd and Liggett's finitely dependent colorings provide earlier one-dependent non-HMM processes [74]. We claim neither first phenomenon. The specific assertion is the simultaneous rank-three, uniformly positive, exactly depolarizing qubit construction and its sharp q, $q+1$, or infinite classification. Classical finite-group recognition, real-algebraic minimization and finite-word equivalence also remain credited [63, 64, 65, 22].

Relation to the causal resource theory. The main article retains the full actual-history comparison theorem and common paid construction. Its lower witness uses the competitor's actual pre-call histories. The paired supplement retains performed-law compactness, variational and dual attainment, robust tied optimality, normalized stopping rigidity, finite-width transfer, rank-constrained discrimination, finite-order quantum measurements, two-stage reset control, memory and communication accounts, spectral geometry and optimal strategy implementation. It also contains the periodic partition, leakage and arithmetic laws, all with their original hypotheses and complete proofs. An independently compiled companion retains absolute calibration and nonintervening directional sensing.

These layers have distinct quantifiers. General causal comparison applies to its declared experiments; the group theorem classifies compactly reversible responses; the posterior theorem supplies a sufficient condition for general instrument families; the exact examples solve their specified models; and local geometric results remain local. The theorem and preservation maps make those dependencies explicit. None of these results is asserted to complete independent A1, A2 or further Theta construction objectives.

2. Prepared kernels, acquired task laws and observers

All hidden-state, action and report spaces are standard Borel. A fixed unknown index $\theta \in \Theta$ specifies an actual preparation and positive normalized causal kernels. Positivity means nonnegative measures, not an everywhere strictly positive density. Dependence on the entire physical past may be included in the hidden state. A legal policy sees past visible reports and its admitted randomization, but not θ or a future audit. Iteration of these kernels defines its finite-prefix laws; the consistent countable path law is their kernel extension.

Each call has a cost $c_i \geq c_* > 0$. The decision to make the next call is predictable. In the geometric route, the declared productive-call flag is also predictable; in the primitive-instrument route, productive reports are specified outcomes and can be random. Commitment is permitted only after a productive report, and occurs exactly once almost surely in each model. At that time a specified future audit $Z \in H$ is scored, where $H \subset \mathbb{R}^k$ is nonempty, compact and convex. The prediction is committed before that audit is observed. A prospective joint report–audit kernel is defined at every productive call, including rejected calls. The stopping coin is independent of the audit conditional on the entire current visible report and the past. Write $L = \operatorname{diam}(H)^2$; the case $L = 0$ is trivial whenever a legal rule exists.

The core cost C_τ includes initial preparation charged to the core, failed calls and all productive calls through commitment. A separately executed final audit must be added to the raw budget. This convention is fixed before taking a risk infimum. Physical time, stored program constants and temporary computation are not identified with call count.

Three resource interfaces. A checkpoint procedure may use its whole precommit history, but must output one element of a fixed dictionary of at most M values. This counts its output width only; it is not a claim of $\log_2 M$ total working bits.

An autonomous observer has an alphabet S, $|S| \leq M$, and one fixed parameter-independent program. At an intercall boundary its only persistent observer memory is its label $s \in S$. Its next command distribution is a fixed function of s. A physical port name, elapsed time, subprogram index, random seed or former report is not an extra free input. During a call its current command, current report and fresh coins are available. Whatever persists to another call must be encoded in S. A fixed decoder $r : S \to H$ reads only the committed label.

In the *event-stopped* interface a post-report kernel emits $(s', e) \in S \times \{0, 1\}$. An event $e = 1$ ends the experiment and retains s' for the decoder; no subsequent command is executed. This public commitment event is not a stored bit used in continuing computation. The same label may be used live on one path and terminal on another. In the *state-stopped* interface a fixed map $h_* : S \to \{0, 1\}$ determines halt status. The initial label is live, and type-one labels are globally absorbing terminal labels with no executed command rows. Entry into a terminal label is allowed only after a productive report. Thus live and terminal types are disjoint in this interface, and only in this interface by definition.

Write $R_i^{\Theta}(N, M)$, $i \in \{\mathrm{cp}, \mathrm{ev}, \mathrm{st}\}$, for the infimum of worst-case expected squared loss over programs with robust almost-sure commitment and $\sup_\theta \mathbb{E}_\theta C_\tau \leq N$. An empty legal class has value $+\infty$. A program chosen once at random is not a free persistent seed. The ideal measurable class may have fixed real constants; effective implementation is stated separately in Section S28.

A physically acquired comparison law. Execute a declared prepared comparison experiment and obtain its joint report–audit probability $K_0(dx, dz)$ on $X \times H$. Its report marginal is ν. The law is genuinely preparable and its specification has its own resource account. It need not be a member of Θ. In particular no finite set of calibration data is silently promoted to a uniform certificate. Set

$$f(x) = \int z K_0(dz \mid x), \quad w(x) = \int \|z - f(x)\|^2 K_0(dz \mid x), \quad d_r(x) = w(x) + \|f(x) - r\|^2. \tag{2.1}$$

Disintegration gives Borel versions with $f \in H$; null sets may be completed arbitrarily. The profile is

$$\mathcal{Q}_m(b) = \min_{\substack{C \subset H,\ 1 \leq |C| \leq m \\ 0 \leq h \leq b,\ \int h\, d\nu = 1}} \int [w + \operatorname{dist}(f, C)^2] h \, d\nu, \qquad b \geq 1,\ m \geq 1. \tag{2.2}$$

We use $+\infty$ for an infeasible cap or width. The quantities f, w are task coordinates of acquired predictive information, not a complete substitute for the predictive quotient. Their derivation and the quotient construction in Appendix S29 precede any use of local dimension.

Lower certificates on actual histories. Fix *one* actual model $\theta_* \in \Theta$, independent of the competing program. At each actual reached productive pre-call history h and legal command a, write $P_{h,a}(dy, dz)$ for its conditional full current report–audit law. Let $P^0_{h,a}$ be a measurable comparison row on the same report space, with a Borel mark $x = \pi_{h,a}(y)$ of law ν and

$$\mathbb{E}^0[Z \mid y] = f(\pi_{h,a}y), \qquad \mathbb{E}^0[\|Z\|^2 \mid y] = w(\pi_{h,a}y) + \|f(\pi_{h,a}y)\|^2. \tag{2.3}$$

These are conditional identities given the *full* report, not just its mark. Versions and inequalities are required on actual reachable histories, not only almost everywhere under an unrelated comparison path law.

Fix $0 < l_- \le 1 \le u_- < \infty$. For all Borel $q : Y \to [0,1]$ and all finite-image Borel $r : Y \to H$, suppose predictable nonnegative defects $a_-(h,a), e_-(h,a)$ obey

$$\int q\, d(P_{h,a})_Y \le u_- \int q\, d(P^0_{h,a})_Y + a_-(h,a), \tag{2.4}$$

$$\int q(y)\|z - r(y)\|^2 P_{h,a}(dy,dz) \ge l_- \int q(y)\|z - r(y)\|^2 P^0_{h,a}(dy,dz) - e_-(h,a). \tag{2.5}$$

Independent current randomization can be adjoined to both rows: integrating inequalities for each fixed random value proves the same inequalities with the same defects. All command-conditioned lower inequalities hold before averaging the chosen command.

Let A_τ count productive calls. At this same actual model require, for every legal competing procedure,

$$\mathbb{E}_* C_\tau \ge \kappa_- \mathbb{E}_* A_\tau, \qquad \kappa_- > 0. \tag{2.6}$$

At budget N suppose, uniformly over all robustly feasible competitors,

$$\mathbb{E}_* \sum_{j \le \tau:\ j \text{ productive}} a_-(H_{j-1}, a_j) \le A_N, \qquad \mathbb{E}_* \sum_{j \le \tau:\ j \text{ productive}} e_-(H_{j-1}, a_j) \le E_N. \tag{2.7}$$

These are actual occupation certificates, not properties merely of the proposed upper program. Uniform row defects give $A_N \le (N/\kappa_-)\bar{a}_-$ and $E_N \le (N/\kappa_-)\bar{e}_-$. Lemma S26.6 supplies a nonuniform alternative.

One common upper construction. Independently, let $\mathcal{G}$ be a parameter-independent program which, from each of its start histories, reaches one productive report. Its conditional expected attempt cost is at most $\kappa_+ > 0$ on every surviving history in every $\theta \in \Theta$. It uses B live roles and returns to its start label on rejection; no reset of the physical state is assumed. It forms a virtual mark x, a current choice J and a permitted export $T_J r$ of its task readout. For every finite dictionary C assume

$$|\{T_j r : j \in J,\ r \in C\}| \le D_0|C|, \qquad D_0 \in \mathbb{N}. \tag{2.8}$$

Every continuing phase and coin needed for this construction is included in the B live roles. For fixed $0 < l_+ \le 1 \le u_+ < \infty$ and $a_+, e_+ \ge 0$, assume for all Borel $q : X \to [0,1]$ and finite-image Borel $r : X \to H$, conditionally on every such history,

$$\mathbb{E}[q(x)] \ge l_+ \int q\, d\nu - a_+, \tag{2.9}$$

$$\mathbb{E}[q(x)\|Z - T_J r(x)\|^2] \le u_+ \int q(x) d_{r(x)}(x)\nu(dx) + e_+. \tag{2.10}$$

The expectation here includes the within-attempt choice J. Unlike the lower, the upper is not required to hold for each choice separately. The choices retained after this call are nevertheless charged through the actual export or live role. These hypotheses express one common causal construction, not modelwise optimizers.

3. Reversible response groups and common controlled memory

The fixed-phase results concern one transition matrix. Several commands require one family of matrices on the same retained labels, and checking every command in isolation is insufficient. We give an intrinsic finite-realization criterion for responses whose minimal linear dynamics is reversible and compact, including any finite family of unitary commands on a prepared finite-dimensional quantum system. We then obtain a matching arithmetic memory law for a pair of noncommuting involutions. This is a theorem about exact controlled response realization, not an assertion that a simulator's private label is an observed sufficient statistic for arbitrary feedback optimization.

Positive realization through invariant polyhedral sets is classical [48, 47, 46]. The argument below uses a section of the realizing simplex before passing to the observable quotient; auxiliary hidden-state responses need not belong to the minimal response space. Compact reversibility then turns containment into equality. The resulting finite-group obstruction concerns all command words, rather than a single spectral restriction.

3.1. **The minimal response space.** Let A and C be nonempty finite alphabets of commands and terminal outcomes. A supplied response is a function $b : A^* \to \Delta(C)$. Its residual after a word w is $b^w(z) = b(wz)$, including the empty word $\varnothing$. Put

$$V_b = \operatorname{span}_{\mathbb{R}}\{b^w : w \in A^*\}, \qquad (L_a f)(z) = f(az).$$

Assume V_b is finite dimensional. It is invariant under each L_a, since $L_a b^w = b^{wa}$. Call the response *compactly reversible* if the restrictions $L_a|_{V_b}$ are invertible and the closure G_b of their generated group is a compact subgroup of $\mathrm{GL}(V_b)$. This concerns the observable quotient; it does not require the full physical command group to be finite or fully observable.

An exact stochastic realization with M live labels has row-stochastic matrices Q_a, one row-stochastic terminal decoder matrix D, and the prescribed initial row e_1^{T}, such that

$$b(a_1 \cdots a_n) = e_1^{\mathsf{T}} Q_{a_1} \cdots Q_{a_n} D. \tag{3.1}$$

Randomized terminal readout is allowed even when all Q_a are deterministic. The target input word is not encoded into an uncounted clock or persistent seed.

Theorem 3.1 (Finite realization of compactly reversible responses)**.** *For a compactly reversible response, the following are equivalent:*

(i) *An exact finite stochastic realization exists.*
(ii) *The residual set $\{b^w : w \in A^*\}$ is finite.*
(iii) *The observable group G_b is finite.*
(iv) *An exact realization with deterministic transitions and a possibly randomized decoder exists.*

The least number of deterministic-transition labels is the residual cardinality. This is not in general the least stochastic label count. If an M-label stochastic realization exists, then

$$|G_b| \le (2^M - 1)!.$$

This last bound is only a finite combinatorial certificate, not a sharp memory law.

Proof. Suppose (3.1) holds, and write $r_w = e_1^{\mathsf{T}} Q_w$. Let $R = \operatorname{span}_{\mathbb{R}}\{r_w\} \subseteq \mathbb{R}^M$. The assignment $\pi(r_w) = b^w$ extends to a well-defined surjective linear map $\pi : R \to V_b$: a relation $\sum_w c_w r_w = 0$ implies, after right multiplication by $Q_z D$, the identical relation among $b^w(z)$ for every z. Moreover

$$\pi(rQ_a) = L_a \pi(r) \qquad (r \in R).$$

The simplex section $K = R \cap \Delta_M$ is a compact polytope, invariant under right multiplication by every Q_a. Its image $P = \pi(K)$ is a compact polytope, contains every residual, and spans V_b. It satisfies $L_a P \subseteq P$. It is important that we intersect the simplex with R first: projecting arbitrary hidden-state responses directly into V_b would not be justified.

For an element T of a compact matrix group, positive powers have a subsequence tending to the identity with exponents tending to infinity. One elementary proof takes a convergent subsequence of powers with successively growing index gaps; quotients of two close powers approach the identity. A finite-order element is covered by its period multiples. Thus positive powers also approach T^{-1}. Applying this to L_a and using closedness of P gives $L_a^{-1}P \subseteq P$, and hence $L_aP = P$. All elements of G_b preserve P. They permute its finitely many vertices, and this permutation action is faithful because those vertices span V_b. Consequently G_b is finite. This proves (i)⇒(iii).

For the displayed bound, a vertex of K is determined by its positive support. If two distinct vertices had the same support, their difference could be added and subtracted in sufficiently small multiples while preserving R, unit mass and nonnegativity, contradicting extremality. Thus K has at most $2^M - 1$ vertices. A linear image of a polytope has no more vertices than the original polytope, so the faithful vertex action proves the bound.

If G_b is finite, the residual orbit is finite. Conversely, if the residual set is finite, each invertible L_a maps it injectively into itself, hence permutes it. Since it spans V_b, the generated group and its closure are finite. This proves (ii)⇔(iii). Under (ii), use the residuals as labels, send b^w to b^{wa} on input a, and use decoder $b^w(\varnothing)$. This realizes b with deterministic transitions. If any such deterministic-transition machine sends u and v to the same label, every suffix yields the same decoder law, so $b^u = b^v$. This proves its minimal count. Finally (iv)⇒(i) is immediate. No assertion that stochastic transitions can always be determinized without increasing size was used. □

Corollary 3.2 (Prepared unitary commands)**.** *Let U_a be finitely many unitary matrices on $\mathbb{C}^d$, let ρ be a prepared density matrix, and let $\{E_r : r \in C\}$ be a terminal positive-operator-valued measurement. For*

$$b(w, r) = \operatorname{tr}(E_r \Phi_w(\rho)), \qquad \Phi_a(X) = U_a X U_a^*,$$

exact finite stochastic realization for all command words is possible if and only if its observable response group is finite. The group is computed after quotienting only those directions invisible to every future commanded terminal test. Its dimension is at most d^2.

Proof. Let $W = \operatorname{span}_{\mathbb{R}}\{\Phi_w(\rho)\}$ in the Hermitian matrices. Each conjugation is a Hilbert–Schmidt isometry. Its forward invariant subspace W is also backward invariant, by finite-dimensional invertibility. Let

$$N = \{X \in W : \operatorname{tr}(E_r \Phi_z X) = 0 \text{ for all } z, r\}.$$

It is invariant under every Φ_a and hence under its inverse. The orthogonal complement of N in W is invariant as well. Therefore the induced transformations on W/N are orthogonal for the quotient inner product. The map $X \mapsto (z, r \mapsto \operatorname{tr}(E_r \Phi_z X))$ identifies W/N with V_b and intertwines them with L_a. Their generated group has compact closure in the orthogonal group. Apply Theorem 3.1. A constant measurement can make this induced group trivial even when the physical unitary group is infinite, so finiteness of the full physical group is not a necessary condition. □

3.2. **A matching memory law for two noncommuting commands.** Fix $x \in \mathbb{R}$ with $\sin(2\pi x) \neq 0$. On a protected qubit take

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} \cos(\pi x) & \sin(\pi x) \\ \sin(\pi x) & -\cos(\pi x) \end{pmatrix}, \quad \rho = E = |0\rangle\langle 0|.$$

Both commands are involutions. For known $0 \leq \alpha \leq 1$, their channels are

$$\Phi_{a,\alpha}(X) = \alpha U_a X U_a^* + (1 - \alpha) \operatorname{tr}(X) I/2.$$

Let $M_x^{\mathrm{ctrl}}(\alpha)$ be the least stochastic live memory of the whole binary response $w \mapsto \operatorname{tr}(E \Phi_{w,\alpha} \rho)$, using the same matrices and decoder for every word. In the real Bloch plane, with coordinates (z, x), these commands are αS_0 and αS_1, where S_0 reflects the angle t to $-t$ and S_1 reflects it to $2\pi x - t$. Their product is $\alpha^2 R_{2\pi x}$.

Theorem 3.3 (All-word arithmetic law for an involutive command pair)**.** *For $0 < \epsilon < 1$ and $\alpha = 1 - \epsilon$, the arithmetic envelope of Section S4 gives*

$$\tfrac{1}{4}\mathfrak{A}_x(\epsilon) \le M_x^{\mathrm{ctrl}}(1-\epsilon) \le (1+4\pi^2)\mathfrak{A}_x(\epsilon). \tag{3.2}$$

The upper is one explicit finite stochastic machine valid for every mixed command word, not separately chosen unary machines. The lower ranges over arbitrary common stochastic realizations.

For irrational x, the lower and upper logarithmic growth exponents as $\epsilon \downarrow 0$ are respectively

$$\frac{1}{\mu(x)+1}, \qquad \frac{\mu(x)-1}{2\mu(x)-1},$$

with values $0, 1/2$ at $\mu = \infty$. A uniform cubic-root law holds precisely for badly approximable x; otherwise the normalized count has lower limit zero and upper limit infinity.

At $\alpha = 1$, the exact count is the reduced denominator q if x is rational, and is infinite otherwise. At $\alpha = 0$, it is two. Every strictly damped pair has a finite exact common realization. These statements concern controlled response memory, not a general multi-command boundary-optimal policy theorem.

Proof. Any common realization restricted to the words $(01)^n$ realizes

$$\tfrac{1}{2}[1 + \alpha^{2n}\cos(2\pi n x)]$$

with transition matrix Q_0Q_1. Its size is therefore at least $m_x(1-\alpha^2) \ge \mathfrak{A}_x(\epsilon(2-\epsilon))/2$ by Theorem S4.1. For $1 \le c \le 2$, the defining maximum of the envelope obeys $\mathfrak{A}_x(c\epsilon) \ge \mathfrak{A}_x(\epsilon)/c$ whenever both arguments are in its domain. This follows for each denominator before taking its infimum. Taking $c = 2 - \epsilon$ proves the lower in (3.2).

For the upper, the rational multiple-polygon construction in the proof of Theorem S4.1 gives a regular l-gon P on the unit circle, with vertex $v_0 = (1, 0)$, such that

$$l \le (1+4\pi^2)\mathfrak{A}_x(\epsilon), \qquad \alpha R_{2\pi x} v_j \in P \quad \text{for every vertex } v_j.$$

The first reflection S_0 permutes this regular polygon and zero belongs to it. Consequently $\alpha S_0 P \subseteq P$, while $S_1 = R_{2\pi x}S_0$ gives $\alpha S_1 P \subseteq P$. Choose barycentric rows $Q_a(j,k)$ of $\alpha S_a v_j$ in the same vertices. Let $d_j = (1 + (v_j)_1)/2$. Starting at vertex zero, induction on the actual command word shows that the expected retained vertex equals the protected Bloch vector. Pairing with the first coordinate proves every response identity. The start is one of the counted vertices; there is no extra initialization mixture.

The two constants in (3.2) transfer both exponents and both uniform-law statements from Theorems S4.3 and S4.4. At zero damping, a rational phase of reduced denominator $q \ge 3$ has a common deterministic q-gon realization. The word-pair transition has a primitive qth-root mode and needs at least q states by the unit-circle spectral obstruction. For irrational x that same obstruction rules out every finite size. At $\alpha = 0$ the response is one at the empty word and $1/2$ after every nonempty word, realized by a first label followed by an absorbing fair label; one label cannot change its decoder. This proves all endpoints. □

A fully algebraic example uses $U_1 = 5^{-1/2}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$. Then $e^{2\pi i x} = (3+4i)/5$ is not a root of unity, since its sum with its conjugate is the rational noninteger $6/5$. Each undamped command in isolation has period at most two; their common all-word response has no finite stochastic realization. The arithmetic exponent of this particular angle is not determined here. The example isolates the failure of commandwise testing without inferring a Diophantine class from finitely many coefficients.

If an independent herald with fixed probability p is placed before each command operation, and only failures apply the supplied command, every common realization above preserves the terminal response under every external command schedule and the first-herald cost $1/p$. The commands are actual inputs to that same retained-state transducer. This does not make its private label observable to the caller, allow uncharged midstream quantum measurements, or prove optimality for a controller

free to choose an easier command sequence. The unary optimal squared-task interpretation below continues to use its separately proved physical embedding.

4. A permutation normal form with an exact memory charge

Theorem 3.1 identifies when finite memory exists. We now determine the form of every possible finite realization, retaining the cost of its prescribed pure initial label. The distinction between a minimal linear representation and a positive realization remains essential. In particular, the permutation representation carried by the observable space does not by itself provide a decoder with values in the probability simplex.

Let b be a compactly reversible response admitting a finite realization. Write G for its finite observable group, $d = \dim V_b$, and N for its residual cardinality. We use the chronological convention $g(u)g(v) = g(uv)$: the right action on V_b is $f \cdot g(w) = L_w f$, where $L_w f(z) = f(wz)$. Thus G in this notation is the opposite of the group of operators with the usual convention for composition; inversion identifies the two groups. No group cardinality, faithfulness, or representation-theoretic statement depends on this convention. Every element has a representative in A^*, including its inverse, since a finite group generated by the command elements is already generated by their positive words. The value $b(g) = b(w)$ is consequently well defined.

A *probability factorization on a finite G-set* consists of a finite set X, row permutation matrices P_g with $P_g P_h = P_{gh}$, a probability row λ on X, and a row-stochastic matrix D with columns indexed by C, such that

$$b(g) = \lambda P_g D \qquad (g \in G). \tag{4.1}$$

All rows of D are probability laws, including rows associated with states that are not distinguished by the immediate terminal measurement. Unused orbits may be deleted. Define

$$\kappa_G(b) = \min\{|X| : (4.1) \text{ holds}\}.$$

The residual realization shows that this minimum is finite and at most N. It permits a mixed λ without charging for its preparation; the stochastic memory $m(b)$ continues to charge every live label, beginning at the specified point mass as in (3.1).

Theorem 4.1 (Reversible stochastic normal form)**.** *Every M-label stochastic realization of b gives a faithful finite G-set X and a factorization* (4.1) *satisfying*

$$|X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \le M. \tag{4.2}$$

Conversely, every such factorization has a realization with the displayed number of labels. Its transitions on X are permutations; if λ is mixed, its only additional label initializes the distribution on the first command. Thus arbitrary continuing stochastic dynamics confers no advantage over one initialization mixture followed by a permutation action.

Proof. Fix a realization $(e_1^{\mathsf{T}}, \{Q_a\}, D_0)$, and retain the reachable row space R and the surjection $\pi : R \to V_b$ from the proof of Theorem 3.1. Write

$$\mathcal{L} = \{w \in A^* : g(w) = 1\}.$$

These are identity words in the observable group; their matrices need not be identity matrices on hidden states. Choose finitely many words $\ell_1, \dots, \ell_s \in \mathcal{L}$ whose positive matrix entries cover all directed edges occurring in any Q_ℓ, $\ell \in \mathcal{L}$. There are at most M^2 distinct edges, so $s \le M^2$ is possible. If no nontrivial choice is needed, the empty word suffices. Set

$$T = \frac{I + Q_{\ell_1} + \cdots + Q_{\ell_s}}{s+1}.$$

Its closed communicating classes $C_1, \dots, C_k$ are closed for every Q_ℓ with $\ell \in \mathcal{L}$. The positive diagonal makes the restriction of T to each such class aperiodic. The elementary convergence argument in

Lemma S2.1 gives a stochastic idempotent

$$E = \lim_{n\to\infty} T^n, \qquad \Delta_M E = \operatorname{conv}\{\mu_1, \dots, \mu_k\},$$

where μ_i is the unique stationary probability row of T on C_i. Their supports are disjoint, so this image is a simplex with exactly k vertices. Every identity-word matrix preserves each C_i; hence

$$EQ_\ell E = E \qquad (\ell \in \mathcal{L}). \tag{4.3}$$

This is a sandwiched equality. We do not assert $Q_\ell E = E$, $EQ_\ell = E$, or common stationarity of the μ_i under all identity words.

Each T^n is a convex combination of identity-word matrices. Consequently $RE \subseteq R$ and $\pi(rE) = \pi(r)$ for $r \in R$. For a word u, put $F_u = EQ_uE$, considered as a stochastic map of $\Delta_M E$ to itself. Choose a word $\bar{u}$ representing $g(u)^{-1}$. Replacing the middle E by the convex combinations T^n and using $u\ell\bar{u} \in \mathcal{L}$ gives

$$F_u F_{\bar{u}} = E = F_{\bar{u}} F_u.$$

Thus F_u is an affine automorphism of that simplex. Such an automorphism permutes its vertices, because both it and its inverse preserve the simplex and therefore preserve extremality. If $g(u) = g(v)$, the same inverse word shows $F_u F_{\bar{u}} = F_v F_{\bar{u}} = E$, so $F_u = F_v$. Next, $EQ_uT^nQ_vE$ is a convex combination of $F_{u\ell v}$, all equal to F_{uv}. Passing to the limit proves $F_uF_v = F_{uv}$. We have obtained a permutation action P_g on the vertices $X = \{\mu_1, \dots, \mu_k\}$.

This action is faithful. If F_u is the identity on $\Delta_M E$, then $rEQ_uE = rE$ for all rows r. For $r \in R$, application of π shows $L_u\pi(r) = \pi(r)$. Surjectivity gives $L_u = I$ on V_b, and so $g(u) = 1$. Write $e_1^{\mathsf{T}} E = \sum_i \lambda_i \mu_i$ and set $D(i, \cdot) = \mu_i D_0$. Intertwining on R gives, for every word w,

$$\lambda P_{g(w)} D = e_1^{\mathsf{T}} EQ_w E D_0 = b(w),$$

including the empty word. If the original initial label lies in C_i, its row under E is exactly μ_i. A mixed λ therefore requires that the initial label lie outside the union of the closed classes. The k nonempty disjoint classes then use at most $M - 1$ labels. This proves (4.2); a pure λ only requires $k \le M$.

For the converse, when λ is a point mass its support point is the initial label in X. If it is mixed, adjoin a label $\star$ with decoder $D(\star, \cdot) = \lambda D$. On command a, its transition row into X is $\lambda P_{g(a)}$, and no transition returns to $\star$. The rows on X are those of $P_{g(a)}$. Induction gives (4.1) for every nonempty word, and the decoder at $\star$ gives the empty word. The first input is applied during initialization, so it is not lost or encoded in an uncounted preparatory clock. □

The general structure of nonnegative idempotents and bounded groups of nonnegative matrices is classical; see Brown [60], Schwarz [61], and Flor [62]. The proof above supplies the additional steps required for an arbitrary semigroup realization: the identity-word projection is constructed before passing to the observable quotient, the sandwiched action is proved to factor through that quotient, and the prescribed pure start is charged. It does not assume that the original stochastic matrices themselves form a group.

Theorem 4.2 (Exact memory formula)**.** *With randomized terminal readout allowed and the initial label prescribed pure, the minimum stochastic memory is*

$$m(b) = \min_{(X,\lambda,D)} \Big(|X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \Big) = \min\{N, 1 + \kappa_G(b)\}. \tag{4.4}$$

The minimum ranges over precisely the probability factorizations (4.1)*, not arbitrary linear factorizations.*

Proof. The first equality is both directions of Theorem 4.1. A point-mass factorization gives a deterministic-transition realization on X, so $|X| \ge N$ by Theorem 3.1. A mixed factorization costs at least $1 + \kappa_G(b)$. Conversely, the residual machine has N labels, and a factorization attaining

$\kappa_G(b)$ costs at most $1 + \kappa_G(b)$. These inequalities give the second equality, also when a minimum factorization happens to have a pure start. □

Corollary 4.3 (Sharp group bounds)**.** *An M-label realization of a compactly reversible response satisfies*

$$|G| \le M!, \qquad M < N \implies |G| \le (M-1)!.$$

Both bounds are attained by families below. In particular, a stochastic memory saving over the residual count requires a faithful permutation core on at most $M-1$ points.

Proof. The core action in Theorem 4.1 embeds G into S_k. If $M < N$, its start cannot be pure, since that would produce a deterministic machine with at most M labels. Thus $k \le M-1$ in this case. □

Let $\nu_G(V_b)$ be the least cardinality of a finite G-set whose real permutation module contains V_b, and let $\mu(G)$ be the least cardinality of a faithful finite G-set. All references to V_b here use the right-action convention already fixed. A probability factorization gives V_b as a subquotient of $\mathbb{R}^X$: take the cyclic span of λ, then its observable quotient. Averaging an inner product over finite G splits every invariant subspace orthogonally, so this subquotient embeds in $\mathbb{R}^X$. We obtain

$$d \le \nu_G(V_b) \le \kappa_G(b), \qquad m(b) \ge \max\{d, \mu(G), \min(N, 1+\nu_G(V_b))\}. \tag{4.5}$$

These are lower bounds, not an identification of a linear module with a bounded decoder. Proposition 4.6 makes the distinction strict.

Corollary 4.4 (Rank rigidity)**.** *For a finite compactly reversible response,*

$$m(b) = d \quad \iff \quad N = d.$$

In that case every realization with d labels has permutation transition matrices. If $N > d$, then $m(b) \ge d+1$.

Proof. Every M-label realization has $d \le M$. If $M = d$, the core in Theorem 4.1 has $k \ge d$ by its linear rank, hence $k = d$ and its start is pure. Thus $N \le d$, whereas $N \ge d$ because the residuals span V_b. Conversely $N = d$ gives a deterministic machine with d labels. For the assertion about the original matrices, a d-label realization has $R = \mathbb{R}^d$ and the observable map π is an isomorphism. An inverse word for $g(a)$ therefore gives a stochastic inverse of Q_a on all of $\mathbb{R}^d$. A stochastic map with a stochastic inverse is an affine automorphism of the full simplex and hence a permutation. □

4.1. **Two commands and a factorial deterministic gap.** Let $k \ge 3$, let P_g be the natural row permutation action of S_k, choose a nonconstant $\lambda \in \Delta_k$, and choose a nonconstant column $t \in [0,1]^k$. Two commands are the k-cycle $(1\,2\,\cdots\,k)$ and the transposition $(1\,2)$; these generate S_k. Supply the binary response

$$b(g) = (\lambda P_g t,\ 1 - \lambda P_g t). \tag{4.6}$$

If the distinct coordinate values of λ have multiplicities $n_1, \dots, n_j$, put $N_\lambda = k!/(n_1! \cdots n_j!)$.

Proposition 4.5 (Exact symmetric-group family)**.** *The response* (4.6) *has observable group S_k, dimension k, residual cardinality N_λ, and*

$$\kappa_{S_k}(b) = k, \qquad m(b) = \min\{N_\lambda, k+1\}.$$

When the coordinates of λ are pairwise distinct, the least stochastic memory is $k+1$ and the least deterministic-transition memory is $k!$. For a point-mass λ, both are k. If either λ or t is constant, the response instead has one residual and one-label memory.

Proof. For any nonconstant vector, differences between it and its transposes yield a nonzero multiple of some $e_i - e_j$. Conjugating by all permutations shows that the span of its orbit differences is $\mathbf{1}^\perp$. The orbit of λ therefore spans $\mathbb{R}^k$, because every row has mass one. The orbit of t also spans $\mathbb{R}^k$: its differences span $\mathbf{1}^\perp$, and its coordinate sum is positive, since a nonconstant nonnegative vector is nonzero. Consequently the map

$$r \longmapsto (h \longmapsto (rP_h t,\, r\mathbf{1} - rP_h t))$$

is injective on $\mathbb{R}^k$. It identifies the reachable row space with V_b, and identifies the observable action with the faithful natural permutation action. Its residuals are precisely the distinct permutations of λ, giving $d = k$ and $N = N_\lambda$. The displayed factorization and the rank lower bound give $\kappa = k$. The memory formula proves the remaining assertion; it also shows $N_\lambda \geq k$. The two constant cases are immediate from (4.6). □

This is an exact prepared quantum experiment in every dimension k: take $\rho = \mathrm{diag}(\lambda)$, permutation unitaries for the commands, and terminal effect $E = \mathrm{diag}(t)$. The construction already lies in the diagonal, classical subalgebra. It demonstrates a stochastic versus deterministic memory separation, and does not assert a quantum advantage for this family. It also proves sharpness of Corollary 4.3: the point-mass family has $M = k$ and $|G| = k!$, while pairwise distinct coordinates have $M = k + 1 < N = k!$ for $k \geq 3$.

For a concrete rational example, take $k = 3$, $\lambda = (1/2, 1/3, 1/6)$, and $t = (0, 1/4, 1)^\mathsf{T}$. The six first-outcome values are

$$\left\{\frac{1}{4}, \frac{3}{8}, \frac{7}{24}, \frac{13}{24}, \frac{11}{24}, \frac{7}{12}\right\}.$$

It has $d = 3$, $N = 6$, $\kappa = 3$, and $m = 4$.

4.2. **The decoder constraint can remove the entire saving.** The representation-theoretic lower bound in (4.5) need not attain the positive factorization minimum, even for the same group, rank, and residual count as this example.

Proposition 4.6 (A strict bounded-decoder obstruction)**.** *Let $u = (1, -1, 0)$ and let S_3 act by its natural permutation matrices. The binary response with first coordinate*

$$\beta(g) = \frac{1}{2} + \frac{1}{4} u P_g u^\mathsf{T} \tag{4.7}$$

has $d = 3$, $N = 6$, and $\nu_{S_3}(V_b) = 3$, but

$$\kappa_{S_3}(b) = m(b) = 6.$$

It is realized by a prepared qubit with

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad \rho = E = |+\rangle\langle +|.$$

Thus the least stochastic memory can differ for responses having the same observable group, module, and residual cardinality.

Proof. The orbit of u consists of its six distinct permutations and spans the two-dimensional zero-sum subspace. Pairing with all permutations of u is nondegenerate on that subspace. Including the constant outcome sum gives dimension three and six distinct residuals. This representation is the natural permutation module $\mathbb{R}^3$, and the rank bound gives $\nu = 3$.

Write $s = (1\,2)$ and $r = (1\,2\,3)$. Directly from (4.7),

$$\beta(1) = 1, \quad \beta(s) = 0, \quad \beta(r) = \beta(r^2) = \frac{1}{4}, \quad \beta(\tau) = \frac{3}{4} \quad \text{for the other two transpositions } \tau.$$

Suppose a probability factorization exists on at most five points. Its action is faithful, by the same residual argument as in Theorem 4.1. Transitive S_3-sets have sizes $1, 2, 3, 6$, since their stabilizers

have orders $6, 3, 2, 1$; the subgroups of order two are conjugate and the subgroup of order three is unique. Thus our faithful action has one natural three-point orbit and, at most, one two-point sign orbit together with fixed points. Two three-point orbits already exceed the budget, and actions on only fixed and sign orbits annihilate the alternating subgroup.

Let a be any point of positive initial mass and write q for the first-coordinate decoder. The endpoint values force $q(a) = 1$ and $q(a \cdot s) = 0$. No fixed point can carry mass. Within the natural orbit, the point fixed by s cannot carry mass, and both of the points exchanged by s cannot carry positive mass simultaneously. Thus the initial mass in that orbit is concentrated at one moved point a; denote its total by A. If a sign orbit is present its mass is concentrated at a single point; write this total as B. We have $A + B = 1$. Every even permutation has decoder one on the mass in the sign orbit, and every odd permutation has decoder zero there.

The images $a \cdot r$ and $a \cdot r^2$ are the other two points in the natural orbit. One is $a \cdot s$, with decoder zero, and the other's decoder is some $t \in [0, 1]$. The two equal values $\beta(r) = \beta(r^2) = 1/4$ give $B = 1/4$ and $At = 0$. Hence $A = 3/4$ and $t = 0$. Of the two transpositions different from s, one fixes a and the other sends it to the third point. Their first-outcome values would therefore be A and zero, whereas both must be $3/4$. This contradiction rules out all factorizations on at most five points. The regular six-point residual machine gives $\kappa \le 6$, so $\kappa = 6$ and Theorem 4.2 gives $m = 6$.

For the physical realization, the adjoint actions of U_0 and U_1 on the real Bloch plane are two reflections whose product is a rotation by $2\pi/3$. The initial and terminal Bloch vector is the x-axis vector. The first reflection sends it to its negative, the other two reflections have inner product $1/2$ with it, and the two nonidentity rotations have inner product $-1/2$. The Born probability is one half of one plus that inner product. It has exactly the displayed six group values, so it is (4.7) after the specified generator assignment. □

The cyclic partition formula of the preceding revision is recovered as a special case in Proposition S2.2. For the general finite group, however, both the start charge and the bounded-decoder constraint belong in the exact minimization problem. Neither can be removed by a rank calculation.

5. Exact memory under terminal readout contrast

The normal form separates the cost of a permutation module from the positivity of its preparation and decoder. We now determine when that linear lower bound is attained, for every finite observable group, and describe the exact memory transitions as the terminal contrast varies. This is a change of terminal readout. It differs from the length-dependent background damping of Section 11.

Let b be a nonconstant compactly reversible response with finite observable group G, residual cardinality N, and real observable module V_b. Set

$$\pi = \frac{1}{|G|} \sum_{g \in G} b(g), \qquad b_t(g) = \pi + t\{b(g) - \pi\}, \qquad 0 \le t \le 1. \tag{5.1}$$

If $\pi_c = 0$, nonnegativity implies $b(g, c) = 0$ for every g; such an output can be removed and later restored as a zero decoder column. Thus positivity of all remaining π_c imposes no additional hypothesis. For prepared unitary commands, (5.1) is realized by the terminal measurement $E_c(t) = tE_c + (1 - t)\pi_c I$, with the preparation and commands unchanged.

Theorem 5.1 (Attainment of the module bound at small contrast)**.** *For every finite observable group G there is an explicitly positive t_0 such that*

$$\kappa_G(b_t) = \nu_G(V_b), \qquad m(b_t) = \min\{N, 1 + \nu_G(V_b)\} \qquad (0 < t \le t_0). \tag{5.2}$$

For every $t > 0$, the observable group, module and residual cardinality agree with those of b. At $t = 0$ the minimum is one label. The assertion allows arbitrary finite groups and arbitrary finite terminal outcome sets.

Proof. Every $f \in V_b$ has a suffix-independent total mass $\ell(f) = \sum_c f(z, c)$, because this holds for its spanning residuals. Group averaging sends every residual to the same constant response $\bar{b}(z) = \pi$. Its image on V_b is therefore $\mathbb{R}\bar{b}$, and

$$V_b = \mathbb{R}\bar{b} \oplus W, \qquad W = \ker\left(\frac{1}{|G|}\sum_{g\in G} L_g\right).$$

The module W has no invariant vector, and $\ell(W) = 0$.

Choose a G-set X of minimum size $k = \nu_G(V_b)$ and an equivariant injection $j : W \to \mathbb{R}^X$. The permutation inner product is invariant. Since group averaging commutes with j, its image is orthogonal to the whole invariant subspace, including the uniform row $p = (1/k, \ldots, 1/k)$. In particular its vectors have coordinate sum zero. Let $F : \mathbb{R}^X \to W$ be orthogonal projection onto $j(W)$ followed by j^{-1}. It is equivariant, $Fj = I_W$, and $F(p) = 0$.

Put $v = b - \bar{b}$, $a = j(v)$, and

$$T(x, c) = F(e_x)(\varnothing, c).$$

The rows of T sum to zero, $pT = 0$, and equivariance gives $aP_gT = b(g) - \pi$. Since $a \neq 0$, we may take

$$\epsilon = \frac{1}{2k\|a\|_\infty}, \qquad B = \max_{x,c}\frac{|T(x, c)|}{\pi_c} > 0, \qquad t_0 = \min\{1, \epsilon/B\}.$$

The row $\lambda = p + \epsilon a$ is a probability row with every entry at least $1/(2k)$. Since $(t/\epsilon)B \leq 1$, for $0 \leq t \leq t_0$ the matrix

$$D_t(x, c) = \pi_c + (t/\epsilon)T(x, c)$$

is row-stochastic. Direct multiplication yields

$$\lambda P_g D_t = \pi + (t/\epsilon)\{pP_gT + \epsilon aP_gT\} = b_t(g).$$

This proves the upper $\kappa_G(b_t) \leq k$ with a bounded decoder.

The linear map $f \mapsto tf + (1 - t)\ell(f)\bar{b}$ is equivariant and invertible for $t > 0$: it is the identity on $\mathbb{R}\bar{b}$ and multiplication by t on W. It sends each residual of b to the corresponding residual of b_t. It preserves the group kernel and the equality pattern among residuals. The module lower bound (4.5) now gives equality, and Theorem 4.2 charges the pure start. At zero contrast the decoder is constant. □

The construction is an equivariant positivity argument. Symmetric cone factorizations and coordinate-symmetric lifts have an established theory [67, 68]; ordinary nonnegative rank also has classical barycentric perturbation results [69]. The assertion here is the exact value $\nu_G(V_b)$ while preserving one common action for every command word and the prescribed-start charge. It is not a claim that positivity under mixing or equivariant factorization was previously unknown.

There is a useful distinction at an infinite compact observable group. The same invertible change of residuals preserves that group for every $t > 0$, so Theorem 3.1 still excludes every finite exact stochastic realization. Thus making the terminal probabilities strictly interior need not make exact memory finite. Terminal readout contrast, repeated channel damping, and perturbations of an entire instrument are different operations.

Theorem 5.2 (Attained exact contrast thresholds)**.** *For each integer $k \geq 1$ there is $\tau_k \in [0, 1]$ such that*

$$\kappa_G(b_t) \leq k \quad \Longleftrightarrow \quad 0 \leq t \leq \tau_k. \tag{5.3}$$

These thresholds are nondecreasing in k, satisfy $\tau_N = 1$, and obey

$$k < \nu_G(V_b) \Longrightarrow \tau_k = 0, \qquad \tau_{\nu_G(V_b)} > 0.$$

The lower memory value is retained at every threshold itself. There are only finitely many positive-contrast transitions. If all $b(g, c)$ are real algebraic, all thresholds are effectively computable real

algebraic numbers. When all subgroups of G are normal, they are obtained by finitely many linear programs; rational tables then have rational thresholds and rational optimal controllers.

Proof. For a fixed action on at most k points, the variables λ, D, t lie in a compact product of probability simplexes and $[0,1]$. The equations $\lambda P_g D = b_t(g)$ define a closed set, so its feasible contrasts form a compact set. If $s > 0$ is feasible, the same λ and the decoder $(t/s)D + (1 - t/s)\mathbf{1}\pi$ make every $0 \le t \le s$ feasible. There are only finitely many actions on at most k labelled points; their feasible sets therefore form a closed initial interval. A single fixed point realizes b_0, including when no positive contrast is feasible.

The residual machine gives $\kappa_G(b_t) \le N$; the remaining inequalities follow from Theorem 5.1 and its module lower bound. The exact memory is recovered from $m(b_t) = \min\{N, 1+\kappa_G(b_t)\}$ for $t > 0$. For algebraic data the equations for each action are finite polynomial constraints over the real algebraic numbers. Classical real quantifier elimination computes their projected interval and endpoint, as in Section 10. This is a proved finite procedure, not a reported general quantifier-elimination execution. The final assertion is the linear reduction and rationality proved in Theorems 6.2 and 6.3 below. □

5.1. **A sharp transition for the retained prepared-qubit experiment.** The bounded-decoder obstruction of Proposition 4.6 has a complete contrast law. Let $u = (1, -1, 0)$ and

$$\beta_t(g) = \frac{1}{2} + \frac{t}{4} u P_g u^{\mathsf{T}}, \qquad g \in S_3,$$

and let $b_t = (\beta_t, 1 - \beta_t)$. This is the same prepared-qubit experiment with terminal effect $t|+\rangle\langle+| + (1-t)I/2$.

Proposition 5.3 (An exact qubit readout threshold)**.** *For $0 < t \le 1$, this response has $d = 3$, $N = 6$, and*

$$\kappa_{S_3}(b_t) = \begin{cases} 3, & 0 < t \le 2/3, \\ 6, & 2/3 < t \le 1, \end{cases} \qquad m(b_t) = \begin{cases} 4, & 0 < t \le 2/3, \\ 6, & 2/3 < t \le 1. \end{cases}$$

At $t = 0$ its memory is one.

Proof. The group, rank and residual count are unchanged for $t > 0$ by Theorem 5.1. A faithful S_3-set of size at most five has exactly one natural three-point orbit and otherwise only fixed points or one two-point sign orbit, as in the proof of Proposition 4.6. Let A be the initial mass on its natural orbit. Write its unnormalized initial row and scalar decoder as

$$p = (A/3)\mathbf{1}^{\mathsf{T}} + r, \qquad D = q\mathbf{1} + s, \qquad r\mathbf{1} = 0, \quad \mathbf{1}^{\mathsf{T}} s = 0.$$

The contribution is $Aq + rP_g s$. All remaining orbits contribute a linear combination of the constant and sign functions. The functions $rP_g s$ and $uP_g u^{\mathsf{T}}$ are orthogonal to both: their group sums vanish because $|S_3|^{-1}\sum_g P_g = \mathbf{1}\mathbf{1}^{\mathsf{T}}/3$, and $\sum_g \operatorname{sgn}(g) P_g = 0$ entrywise. Consequently equality of the full responses implies

$$rP_g s = (t/4) u P_g u^{\mathsf{T}} \qquad (g \in S_3). \tag{5.4}$$

Here no unproved matrix-coefficient injectivity is needed. On the row plane $\mathbf{1}^{\perp}$, use the basis $u, (1, 1, -2)$. The matrices of the identity, (12), (23) and the chronological cycle (123) are respectively

$$I_2, \quad \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} -1 & 1 \\ -3 & -1 \end{pmatrix}.$$

They span all real two-by-two matrices: the first two span the diagonal ones, and the latter two have independent off-diagonal parts. Therefore (5.4) holds with every endomorphism of this plane in place of P_g. Equality of the associated rank-one bilinear functionals, with $t > 0$, forces $r = au$, $s = bu^{\mathsf{T}}$ and $ab = t/4$. Nonnegativity of p gives $|a| \le A/3$, while the bounded decoder gives $|b| \le \min(q, 1-q) \le 1/2$. Hence

$$t/4 = |ab| \le A/6 \le 1/6.$$

No core of size at most five is possible above $t = 2/3$.

For $0 < t \le 2/3$ use the natural three-point action with $\lambda = (2/3, 0, 1/3)$ and scalar decoder $D = \frac{1}{2}\mathbf{1} + (3t/4)u^{\mathsf{T}}$. Its entries lie in $[0, 1]$ and direct multiplication gives β_t; the rank bound proves $\kappa = 3$. Above the threshold the six-residual machine supplies the matching upper. The pure-start formula gives the stated minima and the constant case is immediate. □

6. Quotient decompositions, linear duality and sharp transitions

For groups whose subgroups are normal, the probability optimization in the normal form has an exact linear reduction. This includes every finite abelian group and allows arbitrary probability tables and terminal outcome sets. Neither a special phase choice nor a positive-amplitude trigonometric ansatz is imposed. The reduction uses normal stabilizers; it is not an unrestricted claim that equivariant nonnegative factorization is linear programming.

Lemma 6.1 (Absorption of within-orbit preparation)**.** *Suppose the stabilizers in a finite G-set are normal. Every probability factorization on it can be replaced, without increasing its size, by one having at most one orbit $H\backslash G$ for each normal subgroup H and whose initial row is concentrated at the identity coset within each orbit. Only the mixture between different orbits is required.*

Proof. On an orbit of positive initial mass, normalize its local preparation and write its response as

$$q(g, c) = \sum_{Hx \in H\backslash G} \lambda(Hx) D(Hxg, c).$$

For $h \in H$, normality gives $Hxhg = Hxg$, so $q(hg, c) = q(g, c)$. Thus q itself is a probability decoder on $H\backslash G$. Replacing the preparation by the identity coset and the decoder by q leaves every command-word response unchanged. Orbits with the same H can be merged by averaging their decoders with their initial masses as weights. Delete zero-mass orbits. These operations weakly decrease the size. □

Assume henceforth, until the Fourier specialization, that every subgroup of G is normal. Such groups are called Dedekind groups. For a nonempty set $\mathcal{H}$ of distinct subgroups, put $c(\mathcal{H}) = \sum_{H \in \mathcal{H}} [G : H]$.

Theorem 6.2 (Exact quotient linear programs)**.** *The invariant $\kappa_G(b)$ is the minimum $c(\mathcal{H})$ for which there are $p_H \ge 0$ and functions $f_H : H\backslash G \times C \to \mathbb{R}_{\ge 0}$ satisfying*

$$\sum_c f_H(q, c) = p_H \quad (q \in H\backslash G), \qquad \sum_H p_H = 1, \qquad \sum_H f_H(Hg, c) = b(g, c) \quad (g \in G, c \in C). \tag{6.1}$$

The residual cardinality is $N = |G|$, and $m(b) = \min\{|G|, 1 + \kappa_G(b)\}$. For a rational response table a minimum controller can be chosen rational. More generally, if the table lies in a specified real algebraic number field, the controller can be chosen in that same field.

Proof. Apply Lemma 6.1 to a minimum probability factorization. Its orbit contributions give (6.1). Conversely, if $p_H > 0$, put $D_H = f_H/p_H$, place mass p_H at its identity coset, and use the right quotient action. If $p_H = 0$, all its entries vanish and the orbit is discarded. This constructs the whole response with at most $c(\mathcal{H})$ points. The regular orbit $H = \{1\}$ is always available, and there are finitely many subgroup families.

The stabilizer H_0 of the initial residual is normal. If $h \in H_0$ and $g \in G$, moving h past g using normality shows that h fixes the residual after g. It therefore fixes their span and is the identity in the minimal observable group. Thus $H_0 = \{1\}$ and orbit-stabilizer gives $N = |G|$. The pure-start assertion follows from Theorem 4.2.

For each family the displayed feasible set is a bounded closed polyhedron: $0 \le f_H(q, c) \le p_H \le 1$. A feasible vertex is obtained by solving a linear system over the coefficient field. It is rational for rational data and remains in the given real algebraic field otherwise. The construction uses only

these entries and division by positive p_H. If the residual machine is smaller than the initializer construction, its decoder is the given table and has the same arithmetic property. □

This theorem optimizes the common permutation action supplied by the inherited normal form. The relation between symmetry and positive factorizations is classical [67, 68], and positivity can cost more than a factorization inside the target linear space [70]. The additional step here is the absorption of preparation on normal quotient orbits, which removes the bilinear unknowns without changing the counted resource.

6.1. **An explicit payoff dual.** Let $\mathcal{P}_H$ be the polytope of all probability tables on $H\backslash G$, pulled back to G. For a real table y on $G \times C$, use the unnormalized pairing $\langle y, b\rangle = \sum_{g,c} y(g,c)b(g,c)$ and set

$$h_H(y) = \sum_{q\in H\backslash G} \max_{c\in C} \sum_{g\in q} y(g,c), \qquad h_{\mathcal{H}}(y) = \max_{H\in\mathcal{H}} h_H(y). \tag{6.2}$$

Theorem 6.3 (Payoff certificates and exact contrast endpoints)**.** *The family $\mathcal{H}$ is feasible in* (6.1) *if and only if*

$$\langle y, b\rangle \le h_{\mathcal{H}}(y) \quad \textit{for every real table } y. \tag{6.3}$$

If infeasible, it has a strict rational witness y. A witness of margin $\Delta = \langle y, b\rangle - h_{\mathcal{H}}(y) > 0$ still excludes that family for any table $\widetilde{b}$ with $\|\widetilde{b} - b\|_\infty < \Delta/\|y\|_1$.

For the contrast line (5.1)*, put $v = b - \pi$. Its endpoint for this family is*

$$\tau_{\mathcal{H}} = \min\left\{1, \min_{\langle y,v\rangle=1} \big(h_{\mathcal{H}}(y) - \langle y, \pi\rangle\big)\right\}. \tag{6.4}$$

Both minima are attained. Equivalently, maximize t in $[0,1]$ subject to (6.1) *with right side $\pi + tv$. The endpoint for budget k is the maximum of these linear-program values over $c(\mathcal{H}) \le k$. Rational tables give rational endpoints and optimal controllers; no polynomial bound on the number of families is claimed.*

Proof. The feasible response set for a fixed family is exactly $\mathcal{P}_{\mathcal{H}} = \operatorname{conv} \bigcup_H \mathcal{P}_H$. Each $\mathcal{P}_H$ is a product of probability simplexes. Maximizing a payoff independently on each coset gives its support function h_H; taking a convex hull of the union gives $h_{\mathcal{H}}$. Membership proves (6.3). If b is outside this compact convex set, let q be a closest point. The first variation along the segment from q to any point of the set gives $\langle b - q, x - q\rangle \le 0$. With $y = b - q$ this yields a strictly positive gap of at least $\|b - q\|_2^2$. Approximate y by a rational table sufficiently closely to preserve that gap; the support function is a finite maximum of linear functions and is continuous. The perturbation assertion is the inequality $|\langle y, \widetilde{b} - b\rangle| \le \|y\|_1 \|\widetilde{b} - b\|_\infty$.

The constant table π belongs to every $\mathcal{P}_H$. Along its ray $\pi + tv$, a support inequality with $\langle y, v\rangle \le 0$ imposes no restriction for $t \ge 0$; every other inequality can be normalized by $\langle y, v\rangle = 1$. This gives (6.4). Attainment requires no strict-feasibility assumption. If v lies outside the linear space parallel to the affine hull of $\mathcal{P}_{\mathcal{H}}$, an affine-hull equation gives an attained inner minimum zero. Otherwise the bounded polytope has a finite facet description in that affine hull; the first boundary reached along the ray activates a facet with positive directional coefficient. Its normalized supporting functional attains the inner minimum. The polytope is bounded and $v \ne 0$, so the ray endpoint is finite.

One explicit linear formulation of the inner dual introduces z_{Hq} and s with

$$z_{Hq} \ge \sum_{g\in q} y(g,c) \quad (H, q, c), \qquad s \ge \sum_q z_{Hq} \quad (H \in \mathcal{H}), \qquad \langle y, v\rangle = 1,$$

and minimizes $s - \langle y, \pi\rangle$. The primal formulation is obtained by replacing the right side of (6.1) by $\pi + tv$. It is bounded and contains $t = 0$. Vertex arithmetic proves the endpoint and controller

assertions, and the finite maximum over admissible families gives the budget endpoint. These are exact linear certificates in the specified finite problem, not a formal proof-assistant verification. □

6.2. **Fourier support and a sharp positivity transition.** For an abelian G, let S be the complex characters whose vector Fourier coefficient in b is nonzero. It contains the trivial character and is closed under conjugation.

Corollary 6.4 (The abelian module cover)**.** *The number $\nu_G(V_b)$ is the minimum $\sum_{H\in\mathcal{H}}[G:H]$ over distinct subgroup families such that each character in S is trivial on at least one $H \in \mathcal{H}$. This lower bound is the exact value of $\kappa_G(b_t)$ for all sufficiently small positive contrasts. It need not be attained at full contrast.*

Proof. Character averaging projects the cyclic vector b onto each of its nonzero Fourier modes. Thus its complex cyclic module contains each active character once; conjugate pairs give the corresponding real two-dimensional summands. The permutation module on $H\backslash G$ contains exactly the characters trivial on H, each once. Containment of V_b is therefore exactly the stated cover. Repeated subgroups add no new characters and may be deleted. Apply Theorem 5.1. □

The following example computes a complete rational threshold and an explicit dual obstruction. Let $G = (\mathbb{F}_2)^4$, with the four coordinate flips as commands, and let $\chi_1, \dots, \chi_4$ be its coordinate sign characters. Put $\chi_5 = \chi_1\chi_2\chi_3\chi_4$ and

$$\beta_t(g) = \frac{3}{8} + \frac{t}{8}\sum_{i=1}^{5}\chi_i(g), \qquad b_t(g) = (\beta_t(g), 1-\beta_t(g)), \qquad 0 \le t \le 1. \tag{6.5}$$

Proposition 6.5 (A sharp transition invisible to the observable module)**.** *For every $t > 0$,* (6.5) *has observable group $(\mathbb{F}_2)^4$, dimension six, residual cardinality sixteen, and $\nu_G(V_b) = 10$. Its exact positive core and pure-start memory are*

$$\kappa_G(b_t) = \begin{cases} 10, & 0 < t \le 3/5, \\ 12, & 3/5 < t \le 1, \end{cases} \qquad m(b_t) = \begin{cases} 11, & 0 < t \le 3/5, \\ 13, & 3/5 < t \le 1. \end{cases}$$

At zero contrast the minimum is one. In particular a twelve-label minimum does not occur in this family.

Proof. The five signs have product one, so their sum is $5, 1$, or -3. Consequently β_1 takes the values $1, 1/2, 0$, and every β_t is a valid convex combination with the constant $3/8$. Any four of the five characters are independent over $\mathbb{F}_2$. For $t > 0$ all five character coefficients and the constant mode are nonzero, giving dimension six and trivial joint kernel. Thus G is the observable group and $N = 16$.

For a quotient orbit write $W = H^\perp$ and $r = \dim W$; its size is 2^r. A proper W contains at most r of the five active nontrivial characters, and $2^r \ge 2r$ for $r = 1, 2, 3$. An orbit with $r = 4$ costs sixteen. Covering the five modes therefore costs at least ten, and the five two-point quotients attain this module bound.

Suppose that a positive core has at most eleven points. A size-eight orbit covers at most three active characters; the remaining budget at most three covers at most one more. Size sixteen is unavailable. The nontrivial orbits thus have sizes two or four. Their total cost must be ten and their active-character blocks partition the five modes into singletons or pairs: each dimension must carry an active character and no active character can be covered twice. At most one fixed point can remain. Products belonging to different pairs are distinct and inactive, because equality would give a nontrivial relation involving at most four of the five circuit characters.

For this family of blocks let $\mathcal{B}_2$ denote its pairs and define

$$z(g) = 1 - \sum_{i=1}^{5}\chi_i(g) + \sum_{\{i,j\}\in\mathcal{B}_2}\chi_i(g)\chi_j(g).$$

Averaging z over the stabilizer of a singleton orbit gives $1-\chi_i$; over a pair orbit it gives $(1-\chi_i)(1-\chi_j)$; over a fixed orbit it gives 1. The absence of any relation of length at most four ensures that no other term survives these averages. Each resulting function is nonnegative. Hence any nonnegative quotient contributions satisfy

$$\frac{1}{16}\sum_g z(g)\beta(g) = \sum_H \frac{1}{16}\sum_g z(g) f_H(Hg,1) \geq 0.$$

For the target (6.5), character orthogonality gives the value $(3-5t)/8$. Thus every core of size at most eleven requires $t \leq 3/5$. This is also an explicit instance of Theorem 6.3: take $y(g,1) = -z(g)/16$, $y(g,0) = 0$. Its family support value is zero and its strict margin above the threshold is $(5t-3)/8$.

For $0 < t \leq 3/5$, use five two-point quotient orbits, putting initial mass $1/5$ at the plus point of each. The first-coordinate decoder on orbit i is

$$D_i(g,1) = \frac{3}{8} + \frac{5t}{8}\chi_i(g).$$

Its values lie in $[0,1]$, and the weighted sum is β_t. This attains the ten-point core.

For $t = 1$, use the four-point quotient on (χ_1, χ_2) and the eight-point quotient on (χ_3, χ_4, χ_5), with initial mass $1/2$ at the all-plus point of each. Their decoders are respectively the indicator of two plus signs and the indicator of at least two plus signs. For signs a, b, c, the latter is $1/2+(a+b+c-abc)/4$. The weighted sum of the two decoders is

$$\frac{1+\chi_1+\chi_2+\chi_1\chi_2}{8} + \frac{1}{4} + \frac{\chi_3+\chi_4+\chi_5-\chi_3\chi_4\chi_5}{8} = \beta_1,$$

because $\chi_1\chi_2 = \chi_3\chi_4\chi_5$. Replace each decoder D by $(1-t)3/8 + tD$ for other contrasts. This proves the twelve-point upper everywhere, matching the strict lower above $3/5$. The pure-start formula now gives eleven and thirteen, since $N = 16$. □

Corollary 6.6 (A robust finite-table obstruction)**.** *Fix $3/5 < t \leq 1$ and a binary probability table $\widetilde{\beta}$ on the same group $G = (\mathbb{F}_2)^4$. Supply it under the same four generator commands, so the whole response is $w \mapsto \widetilde{\beta}(g(w))$. If*

$$\max_{g\in G} |\widetilde{\beta}(g) - \beta_t(g)| < \frac{5t-3}{18}, \tag{6.6}$$

its exact positive core has at least twelve points and its pure-start memory is at least thirteen. This is a sufficient lower certificate on finite group tables. It does not assert a sharp neighbourhood radius or a matching thirteen-label upper for every perturbed table.

Proof. Write η for the left side of (6.6). Every normalized Fourier coefficient changes by at most η. Since $(5t-3)/18 < t/8$ for $t \leq 1$, all five prescribed coefficients remain nonzero. Their joint kernel is trivial, so the perturbed table still has observable group G and sixteen residuals. An action of size at most eleven must therefore cover those same five modes, and the preceding proof gives its singleton-and-pair partition even if additional Fourier coefficients are present.

There are zero, one, or two pairs in such a partition. Every permutation of the five circuit characters extends to a linear automorphism of the dual group, because any four are a basis and the fifth is their sum. Thus the absolute sum of its witness z depends only on the number of pairs. Direct evaluation on the sixteen sign patterns gives the following values and multiplicities:

pairs	value : multiplicity of z	$16^{-1}\sum_g \lvert z(g)\rvert$
0	$-4:1,\ 0:10,\ 4:5$	$3/2$
1	$-3:1,\ -1:6,\ 1:4,\ 3:2,\ 5:3$	$17/8$
2	$-2:5,\ 0:4,\ 2:2,\ 4:4,\ 6:1$	$9/4$

The normalized pairing with $\widetilde{\beta}$ is at most

$$\frac{3-5t}{8} + \frac{9}{4}\eta < 0.$$

It cannot have the required nonnegative quotient decomposition. Hence $\kappa_G(\widetilde{b}) \geq 12$, and $m(\widetilde{b}) = \min\{16, 1 + \kappa_G(\widetilde{b})\} \geq 13$. The upper thirteen on the exact contrast line was supplied by a specific construction and is not inferred for arbitrary perturbations. □

For an exact physical realization one may use the sixteen-dimensional diagonal model: prepare the identity basis state, use regular permutation unitaries for the four generators, and take the diagonal terminal effect with entries $\beta_t(g)$. This is already classical; no quantum advantage is asserted for this example. The general low-contrast theorem, the normal quotient linear programs, and this exact transition have different scopes.

7. The retained instrument realization theory and its boundary

The group results concern terminal responses. For an outcome at every call, one must instead preserve each command–outcome history and every subsequent decision made from that visible history. This stronger interface is already covered, under a sufficient uniform posterior condition, by the earlier instrument theory of this paper. We recall its exact scope before proving the new conditional-memory classification.

For fixed known completely positive branches $\mathcal{I}_{a,y}$ whose command sums are trace preserving, the hypothesis is

$$\mathcal{I}_{a,y}(X) \succeq \frac{\eta}{h} \operatorname{tr}(\mathcal{I}_{a,y}(X)) I \quad (X \succeq 0), \qquad \eta > 0.$$

It is a condition on every recorded branch, including the zero-branch convention, and follows when depolarizing noise of weight η is applied after each branch. In every finite dimension $h \geq 2$, and for k prescribed preparations, Theorem S8.2 gives at most

$$k + \left\lceil (4h/\eta)^{h-1} \right\rceil$$

hidden labels, with at most h^2 successors per branch row. All commands use the same physical dictionary. The complete proof in Section S8 gives the operator identity for every history; it preserves all adaptive stopped laws, including nontermination, and nonnegative costs. The hidden label is not exposed to the observer. Corollary S8.3 consequently preserves the risk of every common report-based observer, including constrained boundary budgets; it does not identify hidden simulator size with optimal observer memory.

The same retained theory includes a terminating algebraic construction (Proposition S8.5), interiority after a finite block with the clock and intermediate reports counted (Theorem S8.6), and matching powers of η^{-1} for universal state-positive representation (Theorem S8.4). The latter lower bound is for generators which are actual density matrices. It is not imported as a lower bound against arbitrary nonnegative hidden-state representations. The family in Section 9 will have a lower bound against that unrestricted class.

Invariant positive cones and their stochastic realizations are classical [48, 47]. Okay's quantum simulation framework explicitly includes invariant-polytope updates and adaptive composition [71]. Pure-state nets and approximation of the density-matrix body have an established convex-geometric theory [72]; extension complexity is a different resource [73]. These ingredients, and the paper's already proved interior-instrument theorem, are not asserted as new discoveries of this revision. Their role is to fix the sufficient condition and the resource convention against which the new sharp examples are compared. The next common-contraction construction gives an explicit certificate in supplied linear-response coordinates; it is also an application of positive-realization geometry, not a replacement for the retained instrument theorem.

8. Joint contraction and bounded positive realizations

The reversible classification has a complementary dissipative regime. The condition below is imposed on all command products through one supplied norm, and the construction gives one

machine for all words. A separate realization for each command would not establish that conclusion. Invariant-polytope positive realization is classical [48, 47]; the point of the statement is an explicit sufficient certificate with bounded probability decoders, a prescribed pure start, and a stated number of live labels.

Let A and C be nonempty finite sets, $a = |A|$, and let

$$b(w, c) = p_c + \ell_c(x_w), \qquad x_{wa} = F_a x_w, \qquad x_\varnothing = x_0 \in \mathbb{R}^n, \tag{8.1}$$

where F_a are linear, $p \in \Delta(C)$, $\sum_c \ell_c = 0$, and the displayed values are probabilities for every word. For a norm $\|\cdot\|$ write $\|\ell\|_*$ for its dual norm. Suppose we are given $0 < \rho < 1$ and $r > 0$ such that

$$\|F_a x\| \le \rho \|x\| \quad (a \in A), \qquad r\|\ell_c\|_* \le p_c \quad (c \in C). \tag{8.2}$$

In particular, strictly positive p_c give such an r for every finite collection of output functionals. Zero components are allowed only when the corresponding functional vanishes. The second condition certifies that the entire ball rB, not only the actual orbit, has legal decoder rows.

Theorem 8.1 (A common contracting realization)**.** *Under* (8.2)*, the response* (8.1) *has an exact stochastic realization from a prescribed pure label, valid for every command word, with at most*

$$\sum_{j=0}^{T-1} a^j + \left\lceil \left(1 + \frac{2}{1-\rho}\right)^n \right\rceil \quad \textit{live labels}, \tag{8.3}$$

where $T \ge 1$ may be any integer satisfying $\rho^{T-1}\|x_0\| \le r$. For $x_0 \ne 0$ one may take

$$T = 1 + \max\left\{0, \left\lceil \frac{\log(\|x_0\|/r)}{\log(1/\rho)} \right\rceil \right\};$$

for $x_0 = 0$, the response is constant and one label suffices. No minimality or computational-efficiency assertion is made for this upper bound.

Proof. Put $\varepsilon = 1 - \rho$. Choose a maximal ε-separated finite set V in the unit ball B, where distinct selected points have distance greater than ε. The open balls of radius $\varepsilon/2$ about these points are pairwise disjoint and lie in $(1 + \varepsilon/2)B$. Lebesgue volume, which scales by the nth power, gives

$$|V| \le (1 + 2/\varepsilon)^n.$$

This bound also proves that the greedy selection stops; maximality says that V is an ε-net of B. If $P = \operatorname{conv}(V)$, then $\rho B \subseteq P \subseteq B$. Indeed, for every real linear functional f, choose a maximizer $x \in B$ and a point $v \in V$ at distance at most ε from x. Symmetry of the unit ball gives

$$\max_{v \in V} f(v) \ge f(x) - \varepsilon\|f\|_* = (1 - \varepsilon)\|f\|_*.$$

The separating-hyperplane characterization of convex containment proves $\rho B \subseteq P$. Consequently

$$F_a(rP) \subseteq \rho r B \subseteq rP \quad (a \in A).$$

Use one recurrent label for each $v \in V$, representing the vector rv. For every such label and command choose nonnegative barycentric coefficients of $F_a(rv)$ in rV; they sum to one and define a row of Q_a. Give that label decoder $p_c + \ell_c(rv)$. The decoder is nonnegative by the second condition in (8.2), and its entries sum to one because p has mass one and $\sum_c \ell_c = 0$.

In addition use one transient label for each word w with $|w| < T$, representing its actual vector x_w. Its decoder is the actual probability row $b(w)$, legal by hypothesis. If $|w| < T - 1$, command a sends it deterministically to the label wa. If $|w| = T - 1$, then

$$\|F_a x_w\| \le \rho^T \|x_0\| \le \rho r,$$

so $F_a x_w$ is in rP. Choose its barycentric row in the recurrent labels. Start at the single empty-word label. There are exactly $\sum_{j=0}^{T-1} a^j$ such transient labels before any duplicate states are merged. Induction on the word shows that the probability-weighted represented vector is always x_w: the

transition identity holds on each individual label, and linearity preserves it under a mixture. Pairing with all ℓ_c and adding p gives the entire response. This proves the count, including the pure initial label and every first command. □

The assumption may also be supplied as a uniform product bound. If a known norm $\|\cdot\|_0$ and constants $K \geq 1$, $0 < q < 1$ satisfy

$$\|F_w\|_0 \leq Kq^{|w|} \qquad (w \in A^*),$$

choose any $q < \rho < 1$ and set

$$\|x\|' = \sup_{w \in A^*} \rho^{-|w|} \|F_w x\|_0.$$

This is a norm with $\|x\|_0 \leq \|x\|' \leq K\|x\|_0$ and $\|F_a x\|' \leq \rho\|x\|'$. Hence Theorem 8.1 applies with $\|x_0\|' \leq K\|x_0\|_0$ and any r satisfying $r\|\ell_c\|_{0,*} \leq p_c$. This is a consequence of a supplied uniform certificate, not a decision procedure for joint spectral radius. A common Euclidean contraction certificate, for example, is the finite collection $F_a^{\mathsf{T}} H F_a \preceq \rho^2 H$ for a supplied positive definite H. The actual net and barycentric rows provide finite implementation data; their exact algebraic or sampling encodings remain distinct resources.

9. Average mixing does not control conditional response memory

Positive realization via a common invariant polytope, including composition of adaptive instruments, is established machinery [47, 71]. A hypothesis forcing every normalized branch into an interior region is stronger than fast convergence of the channel obtained by ignoring its outcomes. The following exact example separates these statements even when every outcome has a probability bounded away from zero. The infinite-memory conclusion is a response-level consequence of the inherited relative spectral obstruction; it is not a claim that ordinary mixing theory has failed.

The phase obstruction itself has classical predecessors. The FRDN processes discussed in [50, Section 2.3] already exhibit irrational-phase obstructions to finite classical realization and phase-dependent finite minima in commensurate cases. Our assertion concerns the simultaneous qubit, rank-three, uniform-positivity and exact depolarization properties below, and the sharp additional charge for a counted pure start; it does not claim priority for the phase mechanism.

Fix $\sin\theta \neq 0$. For a qubit put

$$U_\theta = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad E = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad \tau = |0\rangle\langle 0|, \qquad \mathcal{D}(X) = \operatorname{tr}(X) I/2.$$

Define three linear maps

$$\Psi_0(X) = \tfrac{1}{8} U_\theta X U_\theta^*, \quad \Psi_1(X) = \tfrac{1}{4} \operatorname{tr}(EX)\tau, \quad \Psi_2 = \mathcal{D} - \Psi_0 - \Psi_1. \tag{9.1}$$

Theorem 9.1 (Exact memory in a uniformly positive one-dependent family)**.** *The maps* (9.1) *form a quantum instrument. Its unconditioned channel is the completely depolarizing map $\mathcal{D}$. For every density matrix, its three outcome probabilities belong respectively to*

$$\{1/8\}, \qquad [1/16, 3/16], \qquad [11/16, 13/16].$$

Started at $I/2$, its stationary outcome process is one-dependent: two finite blocks separated by at least one unobserved outcome are independent. Its word-probability Hankel matrix has rank exactly three. Write $z = e^{i\theta}$. If z has finite order q, the least classical hidden-state count reproducing every word probability is exactly q when a mixed initial row is free, and exactly $q+1$ when the initial label must be pure and counted. If z is not a root of unity, no finite classical model exists, even with a freely mixed start and arbitrary real stochastic entries. All primitive superoperators are algebraic whenever $\cos\theta$ and $\sin\theta$ are algebraic.

Proof. Use the Choi convention $J(\Phi) = \sum_{i,j=0}^{1} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|)$. Then

$$J(\mathcal{D}) = \tfrac{1}{2} I_4, \qquad \|J(\Psi_0)\|_{\mathrm{op}} = \tfrac{1}{4}, \qquad \|J(\Psi_1)\|_{\mathrm{op}} = \tfrac{3}{16}.$$

The first two branch Choi matrices are positive semidefinite. Subtraction therefore gives $J(\Psi_2) \succeq I_4/16$, proving complete positivity. The sum is trace preserving by construction. The probability bounds follow from $\operatorname{tr} \Psi_0(\sigma) = 1/8$ and $\operatorname{tr} \Psi_1(\sigma) = \operatorname{tr}(E\sigma)/4 \in [1/16, 3/16]$. Conjugation by U_θ depends algebraically on $\cos\theta$ and $\sin\theta$, which proves the last assertion.

The initial state $\sigma_* = I/2$ is fixed by $\mathcal{D}$, giving stationarity. To check one-dependence directly, write Ψ_u for the chronological branch product of a block u. Summing over one intervening outcome gives

$$\sum_{s=0}^{2} \operatorname{tr}\big(\Psi_v \Psi_s \Psi_u(\sigma_*)\big) = \operatorname{tr}\big(\Psi_v \mathcal{D} \Psi_u(\sigma_*)\big) = \operatorname{tr} \Psi_u(\sigma_*)\ \operatorname{tr} \Psi_v(\sigma_*).$$

Further unobserved outcomes leave $\mathcal{D}$ unchanged. Summing over specified block events proves the claimed independence.

After outcome 1 the normalized state is τ, and that first outcome has probability $1/8$. Repeating outcome 0 rotates this state with the factor 8^{-n}. The probability of the full word $1\,0^n\,1$ is

$$u_n = \frac{1}{64}\, 8^{-n} \left(1 + \frac{1}{2}\cos(n\theta)\right). \tag{9.2}$$

All reachable subnormalized states are real symmetric matrices, a space of real dimension three, so the word-probability Hankel rank is at most three. For the reverse inequality take prefixes $1\,0^i$ and suffixes $0^j\,1$, $0 \le i, j \le 2$. Their Hankel submatrix is (u_{i+j}). Equation (9.2) expresses it as the product of a three-point Vandermonde matrix, a diagonal matrix with three nonzero coefficients, and the transpose Vandermonde matrix. It is therefore nonsingular, proving rank three.

Suppose a finite classical model exists. Write its nonnegative outcome matrices as T_0, T_1, T_2, with $\sum_s T_s$ row-stochastic, and its initial probability row as μ. Then

$$u_n = v T_0^n d, \qquad v = \mu T_1 \ge 0, \quad d = T_1 \mathbf{1} \ge 0.$$

Here T_0 is row-substochastic. Delete states unreachable from the support of v and states from which no positive terminal value of d is reachable. The remaining principal submatrix S still realizes u_n. For every remaining i, j, choose a positive path from v to i, of length a_i and weight $p_i > 0$, and a path from j to a positive terminal value, of length b_j and weight $r_j > 0$. Nonnegativity gives

$$p_i r_j (S^n)_{ij} \le u_{a_i+n+b_j} \le C_{ij} 8^{-n}.$$

Thus $(8S)^n$ is entrywise bounded, and every eigenvalue of S has modulus at most $1/8$.

The three exponential modes of (9.2) are $1/8$, $z/8$, and $\overline{z}/8$, all distinct and with nonzero coefficients. Cayley–Hamilton and the Vandermonde independence of these sequences force $z/8$ to be an eigenvalue of S. In an upper block triangular decomposition of S, it belongs to an irreducible nonnegative diagonal block B. Its spectral radius is exactly $1/8$. Perron normalization by a strictly positive eigenvector therefore makes $8B$ similar to a stochastic matrix with eigenvalue z. This is the elementary normalization proved before Proposition S6.2. At a largest-modulus coordinate of a corresponding complex eigenvector, equality in the triangle inequality forces each positive successor to have the same modulus and value multiplied by z. Irreducibility propagates this equality to every coordinate, so the relation holds on every positive edge and every directed cycle. Following positive successors reaches a directed cycle; its length $l \ge 1$ gives $z^l = 1$. This identity proves that z must be a root of unity. If its order is q, every such directed cycle has length divisible by q, so every realization has at least q labels.

For the matching upper, suppose z has order $q \ge 3$. In the real Bloch plane let P be the regular q-gon with vertices

$$r_j = (\cos(2\pi j/q), \sin(2\pi j/q)), \qquad 0 \le j < q,$$

and let σ_j be the corresponding pure qubit states. The polygon contains the disk of radius $\cos(\pi/q) \geq 1/2$. The normalized branch 0 permutes its vertices and branch 1 resets to $\sigma_0 = \tau$. For an arbitrary state with Bloch vector r in the unit disk, write $h = \operatorname{tr} \Psi_1(\sigma) \in [1/16, 3/16]$. The normalized Bloch vector after branch 2 is

$$\frac{-R_\theta r/8 - h r_0}{1 - 1/8 - h},$$

whose norm is at most $(1/8 + 3/16)/(11/16) = 5/11 < 1/2$. It therefore lies in P. For each input vertex and outcome, expand the normalized output in the same vertices and multiply the barycentric coefficients by its outcome probability. These nonnegative rows form outcome matrices T_s with $\sum_s T_s$ row-stochastic. They reproduce each subnormalized branch state on each label. Induction proves equality of every word probability. The uniform mixture of the vertices represents $I/2$, giving the q-label upper with a free mixed start. A stationary hidden initialization can also be used: with $T = \sum_s T_s$, every row μT^n represents $\mathcal{D}^n(I/2) = I/2$, where μ is the uniform row. A convergent subsequence of their Cesaro averages gives $\pi T = \pi$, and π still represents $I/2$, so it gives the same word probabilities on q labels. If a pure start is required, add one label representing $I/2$ and use its branch decompositions on the first outcome; the first outcome is generated during this counted transition. No transition returns to that label. This gives the $q + 1$ upper.

It remains to exclude a pure-start realization on exactly q labels. The preceding lower then forces the irreducible block carrying $z/8$ to use every label. Perron normalization makes $8T_0$ similar to a stochastic matrix with primitive qth-root eigenvalue and exactly q states. The equality argument above supplies a directed q-cycle using every state. An additional positive edge would close a directed cycle of length less than q, which is impossible. Thus, before normalization, T_0 is a weighted directed q-cycle. Its successive row weights are positive. From the prescribed starting point,

$$p(0^n) = 8^{-n} \qquad (n \geq 0).$$

For $n = 1, \ldots, q$, the left side is the product of the first n row weights encountered on this cycle. Successive ratios force every weight to be $1/8$, so $T_0 = C_q/8$ after relabelling. Next, $p(0^n 1) = 8^{-n}/8$ shows that every coordinate of $d = T_1 \mathbf{1}$ is $1/8$. This would give

$$p(1\, 0^n\, 1) = e_1^{\mathsf{T}} T_1 (C_q/8)^n d = 8^{-n}/64,$$

contradicting (9.2), already at $n = 0$. The pure-start minimum is therefore $q + 1$. □

In particular, the fully algebraic choice

$$U_\theta = 5^{-1/2} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad e^{i\theta} = (3 + 4i)/5,$$

has no finite classical realization. Its superoperator coefficients are rational. The phase is not a root of unity, since otherwise $z + z^{-1} = 6/5$ would be a rational algebraic integer and hence an integer. Every finite-order choice, on the other hand, gives the exact counts q and $q + 1$ in a fixed physical qubit with the same uniform outcome probability bounds. The size of an exact hidden-state realization cannot be inferred from predictive rank, unconditional mixing time, or finite dependence range.

Stationary one-dependent processes without a finite hidden Markov model were established earlier, notably by the finitely dependent colorings of Holroyd and Liggett [74]. We make no first-existence claim for that phenomenon. Here the simultaneous properties are an explicit qubit instrument, rank-three linear prediction, uniformly positive outcome probabilities, and an exactly depolarizing unconditioned channel, together with a sharp finite-order memory law and a separate pure-initialization charge.

This example satisfies a stronger average-mixing property than exponential convergence: one unobserved step erases all prior state information. The infinite obstruction survives in a specified observed word, whose conditional maps retain an irrational phase. It therefore does not contradict

a finite-realization theorem requiring uniform mixing of each normalized branch. It also explains why positive single-outcome probabilities cannot replace that hypothesis. The example supplies a precise boundary for the constructive theorem.

10. Exact synthesis from algebraic observable data

The preceding normal form concerns existence over the real numbers. We now give a terminating exact procedure on algebraic input, including a certificate that no finite stochastic realization exists. The input is a supplied finite-dimensional probability response, rather than a finite sample from an unknown process. All algebraic numbers are specified by defining polynomials and isolating data, with a common real number field and its chosen embedding. The algorithms below concern exact label count and exact controller tables. They make no polynomial running-time or bounded-bit sampling claim.

Finite matrix-group recognition is classical, and substantially faster algorithms than the elementary enumeration below are known [64, 63]. Reduction of probabilistic-automaton minimization to the existential theory of the reals is also established [65]. Together with the inherited reversible criterion, these ingredients already yield decidability and bounded-size minimization on the input class considered here. The new normal form supplies the more structured conclusion: a smallest controller can be returned as a finite permutation core with a precisely counted initializer and a bounded multi-output decoder. We give a complete terminating construction, including finite certificates of the negative answer. Neither finite-group recognition nor real-algebraic automaton minimization is asserted to be a new algorithm.

10.1. **Observable reduction and a finite stopping bound.** Write the supplied representation in chronological row convention as

$$b(w) = \beta T_w B, \qquad T_{uv} = T_u T_v, \tag{10.1}$$

where the matrices, row and decoder have entries in a real number field K of degree e over $\mathbb{Q}$. The displayed representation is linear; its matrices and intermediate coordinates need not be stochastic or nonnegative. Probability validity means that the supplied $b(w)$ belongs to $\Delta(C)$ for every word. Compact reversibility is imposed on its *minimal observable* response space, as in Section 3.

For clarity, minimal reduction itself is finite. Starting with β, close its row span under right multiplication by each T_a. Starting with the columns of B, close their span under left multiplication by the T_a. Either iteration stabilizes once its dimension fails to increase, and neither dimension exceeds the supplied dimension. A reachable row is invisible precisely when it annihilates the second span. Quotienting the reachable span by this kernel gives the minimal representation, with dimension $d = \dim V_b$ and coefficients still in K. This is the familiar reachable/observable linear reduction; it does not identify arbitrary hidden stochastic rows with observable states. Since the response is normalized and nonempty, $d \geq 1$.

Set

$$h = ed, \qquad B_h = |\operatorname{GL}_h(\mathbb{F}_3)| = \prod_{j=0}^{h-1} (3^h - 3^j). \tag{10.2}$$

This deliberately conservative bound has the advantage of a short complete proof. Restriction of scalars may use a larger field than necessary; doing so changes the computational bound, not the resulting exact label minimum.

Lemma 10.1 (Classical finite-group bound)**.** *A finite subgroup of* $\operatorname{GL}_d(K)$ *has order at most* B_{ed}*. Consequently a monoid generated by invertible* d*-dimensional matrices over* K *is finite if and only if breadth-first enumeration of its distinct positive-word products closes before producing* $B_{ed} + 1$ *elements.*

Proof. Restriction of scalars embeds the group faithfully in $\mathrm{GL}_{ed}(\mathbb{Q})$. For a finite subgroup H of $\mathrm{GL}_h(\mathbb{Q})$, let $\Lambda = \sum_{g\in H} g\mathbb{Z}^h$. There is a common denominator q with $\mathbb{Z}^h \subseteq \Lambda \subseteq q^{-1}\mathbb{Z}^h$. Thus Λ is a free abelian group of rank h, and it is invariant under every g and g^{-1}. In a lattice basis, H is a subgroup of $\mathrm{GL}_h(\mathbb{Z})$.

The kernel of reduction modulo 3 has no nonidentity element of finite order. Indeed, such an element would have a power A of prime order ℓ. Write $A = I + 3^s Z$, with $s \geq 1$ and with an integer matrix Z not zero modulo 3. The binomial expansion of $A^\ell = I$, divided by 3^s, gives $\ell Z = 0$ modulo 3, so $\ell = 3$. The expansion for this prime, divided by 3^{s+1}, instead gives

$$Z + 3^s Z^2 + 3^{2s-1} Z^3 = 0,$$

again contradicting $Z \not\equiv 0 \pmod 3$. Reduction is therefore injective on H. Counting independent ordered columns over $\mathbb{F}_3$ proves the bound in (10.2). This is the standard Minkowski argument [63]; the proof is included to make the termination certificate explicit.

A finite monoid of invertible matrices is a group: for every element, two positive powers agree, and invertibility yields a positive power equal to the identity and another equal to its inverse. A finite generated monoid therefore has at most B_{ed} elements. In a breadth-first search, process each newly encountered product and multiply it by all generators. If the queue empties, the finite set found is closed under all generators and is the entire monoid. If the monoid is infinite, the search eventually finds $B_{ed} + 1$ distinct products. These are the two asserted stopping events. All equality tests are exact tests in K. □

Theorem 10.2 (Decidability and minimal algebraic synthesis)**.** *For a supplied compactly reversible probability response with algebraic linear representation* (10.1)*, there is a terminating algorithm that decides whether an exact finite stochastic realization exists. If the answer is positive, the algorithm returns its least number $m(b)$ of live labels and an exact realization with algebraic entries, a prescribed initial label and a row-stochastic terminal decoder. It can return a permutation core with at most one transient initializer, as in Theorem 4.1.*

A negative answer has a finite certificate consisting of $B_{ed} + 1$ distinct observable command products. A positive answer has an exact controller, the finite observable-group multiplication table, and finite real-algebraic feasibility or infeasibility certificates for every smaller candidate size. The negative certificate concerns the promised compactly reversible input class; it is not a nonrealizability test for arbitrary dissipative responses.

Proof. Perform the observable reduction and apply Lemma 10.1 to the minimal command matrices. If enumeration exceeds its bound, the positive-word monoid and its compact group closure are infinite. Theorem 3.1 rules out every finite stochastic realization. The list of distinct products, each supplied with its generating word, is a finite independently checkable certificate of this conclusion.

Otherwise enumeration closes to a finite group G. Compute the distinct reachable rows βT_g in the minimal representation; their count n is the residual cardinality. The deterministic residual machine proves $m(b) \leq n$. Theorem 4.2 expresses the answer as a minimum over finite permutation actions, a probability row λ, and a stochastic decoder D. Only the following finite possibilities can improve or attain the upper n: a core of size $k \leq n$ with a point initial row, or a core of size $k \leq n - 1$ with a possibly mixed initial row and one additional initializer.

For each such k, enumerate assignments of permutations to the elements of G and retain those satisfying $P_1 = I$ and $P_g P_h = P_{gh}$. This is a finite search through at most $(k!)^{|G|}$ assignments, and it may of course be reduced using generators and relations. For a retained action impose

$$\lambda_i \geq 0, \quad \sum_i \lambda_i = 1, \qquad D_{ic} \geq 0, \quad \sum_c D_{ic} = 1, \qquad \lambda P_g D = b(g) \quad (g \in G). \tag{10.3}$$

For the point case impose $\lambda = e_i^{\mathsf{T}}$ and enumerate i. For the other case it is harmless to permit point rows as well: that candidate then pays a redundant initializer, and the separately enumerated point case can only improve its cost. Equations (10.3) have algebraic coefficients and degree at most two

in their real unknowns. Exact real-algebraic elimination decides feasibility, and a nonempty such set has an algebraic sample point. One may see the latter directly from a sign-invariant algebraic cell decomposition: choose isolating algebraic sample coordinates successively in its nonempty sections and sectors. Algebraic elimination is the classical decision theorem used here [15]; no numerical optimizer is being substituted for it.

Search in increasing total label cost and stop at the first feasible candidate. Termination follows from the deterministic upper. Theorem 4.2 proves that no stochastic competitor outside the enumerated normal forms can be smaller. For a point initial row use the permutation matrices directly. For a mixed initial row add the counted initial state, give it decoder $b(\varnothing) = \lambda D$, and on command a send it to λP_{g_a} in the core. All later transitions are permutations. Every returned entry is algebraic, and all response identities follow from the group action and (10.3).

The finite word criterion of Theorem 11.1 supplies an alternative bounded-size formulation using arbitrary stochastic matrices, independently of the normal form. This is useful for checking a returned implementation against the original linear representation. Its existence does not give an efficient algorithm for minimizing probabilistic automata, a problem with established hardness results [65]. □

Corollary 10.3 (Algebraic prepared unitary experiments)**.** *Let the prepared density matrix, a finite family of unitary commands and a finite terminal positive-operator-valued measurement have algebraic real and imaginary entries. Whether their Born probabilities for all command words admit an exact finite classical stochastic realization is decidable. In the positive case a smallest exact classical realization and its bounded decoder can be synthesized with algebraic entries. For a protected Hilbert space of dimension d_{phys}, the observable linear dimension used in the algorithm is at most d_{phys}^2.*

Proof. Hermitian matrices have a real coordinate basis of size d_{phys}^2 consisting of diagonal matrix units and real and imaginary symmetric matrix-unit pairs. In this basis conjugation by each supplied unitary, preparation, and terminal evaluation have real algebraic coefficients. This constructs (10.1). Positivity and normalization of Born probabilities follow from the declared density matrix and measurement, not from a finite test. The observable reduction is finite linear algebra over a common real number field. Corollary 3.2 proves compact reversibility of this quotient, so Theorem 10.2 applies. □

This corollary allows noncommuting commands, nonfaithful physical actions, mixed preparations and arbitrary finite terminal measurements. In particular, a physically infinite group with constant observable response is correctly reduced to the one-label case before group enumeration. The result concerns exact simulation of the supplied experiment. Choosing an optimal feedback experiment, deciding validity of an arbitrary algebraic series, approximate program size, exact sampling and workspace lower bounds are different questions. The complete retained causal and boundary theory keeps its own quantifiers and resource accounts.

11. Finite certificates and stable exact memory

The normal form gives exact realizations at reversible responses. We next prove that their memory obstructions admit finite tests and persist under a specified class of physical perturbations. Finite-word equivalence for linear machines is classical [22]; the finite bound below is its direct difference-space argument. The additional conclusions concern the positive separation from *all* smaller stochastic machines, its effective algebraic certificate, and a matching realization under damping toward a common background response.

11.1. **Finite tests against a fixed memory budget.** Suppose a response $b : A^* \to \Delta(C)$ has a supplied real linear representation

$$b(w) = \beta T_w B, \qquad \beta \in \mathbb{R}^{1\times d}, \quad T_a \in \mathbb{R}^{d\times d}, \quad B \in \mathbb{R}^{d\times|C|}.$$

Minimality of this representation is not required. For $M \geq 1$ let $\mathcal{P}_M$ be the compact product of the row-stochastic matrix simplexes for $(Q_a)_{a\in A}$ and $D \in \mathbb{R}^{M\times|C|}$. Its initial row is the prescribed pure row e_1^{T}, as before. Set

$$L = d + M - 1, \qquad \delta_M(b) = \min_{(Q,D)\in\mathcal{P}_M} \max_{\substack{|w|\leq L\\ c\in C}} \left| e_1^{\mathsf{T}} Q_w D_c - b(w,c) \right|. \tag{11.1}$$

Theorem 11.1 (Finite obstruction and effective separation)**.** *An M-label stochastic machine realizes b on every word if and only if it does so on the words of length at most $d + M - 1$. Consequently $\delta_M(b) > 0$ if and only if no exact M-label realization exists. If another supplied response $\widetilde{b}$ satisfies*

$$\max_{|w|\leq L,\ c\in C} |\widetilde{b}(w,c) - b(w,c)| \leq \eta,$$

then every M-label machine has error at least $\delta_M(b) - \eta$ on these same probes. This last assertion imposes no finite-rank hypothesis on $\widetilde{b}$.

If the given entries of β, T_a, B are real algebraic numbers, $\delta_M(b)$ is an effectively computable real algebraic number, and a minimizing machine with real algebraic entries can be computed. When it is positive, a positive rational number strictly below it and a real-algebraic quantifier-elimination certificate of that strict lower bound can be computed. No polynomial-time or practical-size bound is asserted.

Proof. For a fixed candidate machine use the block-diagonal matrices $S_a = \operatorname{diag}(T_a, Q_a)$, the row $u = (\beta, -e_1^{\mathsf{T}})$, and the output columns $v_c = (B_c, D_c)^{\mathsf{T}}$. The difference on w is $uS_w v_c$. Let R_j be the span of uS_w for $|w| \leq j$. These spaces increase; if $R_j = R_{j+1}$, the common space is invariant under every S_a and all later spaces coincide. Since $u \neq 0$, the sequence starts in dimension one inside a space of dimension $d + M$ and must have stabilized by $j = d + M - 1$. Vanishing on the indicated finite set therefore implies vanishing on all words. This is the same linear-equivalence principle as the inherited finite-realization test.

The maximum in (11.1) is a continuous function on the compact set $\mathcal{P}_M$, so its minimum is attained. It is zero precisely when one machine passes every finite probe, hence, by the first assertion, realizes every word. The perturbation claim follows pointwise from the triangle inequality and then from the minimum defining $\delta_M(b)$.

For algebraic input, every probe value is algebraic. The constraints defining row-stochasticity and the inequalities between the finitely many polynomial errors and a variable t are polynomial constraints over the real algebraic numbers. Real quantifier elimination decides their feasibility and computes the least feasible t together with an algebraic witness. A singleton semialgebraic set over the real algebraic numbers consists of an algebraic number. If $\delta_M(b) > 0$, effective isolation supplies a positive rational $\sigma < \delta_M(b)$; infeasibility with all errors at most σ is a finite real-algebraic certificate. This uses the classical decision procedure [15]; it does not assert that this procedure was executed by the finite diagnostics accompanying this article. □

For the prepared quantum response, these finite probes have an explicit physical perturbation bound. Suppose two density matrices differ in trace norm by at most h_ρ, corresponding channels differ in diamond norm by at most h_Φ, and corresponding terminal effects differ in operator norm by at most h_E. All channels are trace preserving and completely positive, and both lists of effects are valid measurements. Contractivity and telescoping give $\|\widetilde{\Phi}_w(\widetilde{\rho}) - \Phi_w(\rho)\|_1 \leq h_\rho + |w|h_\Phi$. The state difference has trace zero; subtracting $I/2$ from a terminal effect, whose spectrum lies in $[0,1]$, gives

$$\max_{|w|\leq L,c} |\widetilde{b}(w,c) - b(w,c)| \leq h_E + \tfrac{1}{2}(h_\rho + Lh_\Phi). \tag{11.2}$$

Thus a strict smaller-memory obstruction survives a specified open neighbourhood of physical primitives. This supplies a lower bound there; it does not assert that an arbitrary perturbation has a realization of the same size.

11.2. **A matching upper under a common background.** Let b be compactly reversible, nonconstant, and have finite stochastic memory $m = m(b)$. Let G be its finite observable group, with the chronological convention of the normal-form theorem, and put

$$\overline{b} = |G|^{-1} \sum_{g \in G} b(g), \qquad C_b = \max_{g \in G, c \in C} |b(g, c) - \overline{b}(c)| > 0.$$

Choose a damping factor $\alpha_a \in [0, 1]$ for each command, put $\beta_w = \prod_{j=1}^{|w|} \alpha_{w_j}$, with $\beta_\varnothing = 1$, and define

$$b_{\boldsymbol{\alpha}}(w) = \beta_w b(w) + (1 - \beta_w)\overline{b}. \tag{11.3}$$

Theorem 11.2 (Stable exact memory under background damping)**.** *The response* (11.3) *has an exact realization with m labels for every choice of the damping factors. Put $L = d + m - 2$, where d is any supplied representation dimension of b, and let $\epsilon = \max_a(1 - \alpha_a)$. Then*

$$\epsilon < \min\left\{1, \frac{\delta_{m-1}(b)}{LC_b}\right\} \quad \Longrightarrow \quad m(b_{\boldsymbol{\alpha}}) = m. \tag{11.4}$$

For algebraic input satisfying the hypotheses, a positive algebraic cutoff, or a smaller positive rational cutoff, is effectively computable. For a constant response every damping leaves that response unchanged and its minimum is one label; the expression involving $m - 1$ is not used.

Proof. Take a minimal realization in the normal form proved above. On its recurrent labels each Q_a is a permutation, and an extra initializer is present only when the initial distribution on these labels is mixed. Average that distribution over G, and call the resulting row π, extended by zero on the initializer when present. It obeys

$$\pi Q_a = \pi, \qquad \pi D = \overline{b}.$$

Indeed, the group average is invariant under each permutation. The initializer has no incoming transition, and its first-input row is the original initial distribution followed by the commanded permutation, so the same identities hold on the entire machine. Set $J = \mathbf{1}\pi$. Then $J^2 = J$, $Q_a J = J Q_a = J$, and

$$Q_{a,\boldsymbol{\alpha}} = \alpha_a Q_a + (1 - \alpha_a) J$$

is row-stochastic on the same labels. Multiplication gives, for every nonempty word and also trivially at the empty word,

$$Q_{w,\boldsymbol{\alpha}} = \beta_w Q_w + (1 - \beta_w) J.$$

Reading with the same decoder proves the upper.

By minimality $\delta_{m-1}(b) > 0$. For $|w| \le L$, the elementary product bound $1 - \beta_w \le |w|\epsilon$ gives an error from b of at most $LC_b\epsilon$. Theorem 11.1 rules out every $(m - 1)$-label realization when this is smaller than $\delta_{m-1}(b)$. Machines with fewer labels can be padded by unreachable rows, so they too are excluded. This and the upper prove equality. A nonconstant response has $m \ge 2$ and $d \ge 2$, hence $L > 0$. The effective assertion follows from the finite group enumeration and algebraic synthesis theorem, together with Theorem 11.1. Finite maxima and averages over the computed group preserve algebraicity. □

There is a physical realization of the common background for prepared unitary commands. Let $\mathcal{K}$ be the compact closure of their physical conjugation group. The compact convex hull of $\{\Phi(\rho) : \Phi \in \mathcal{K}\}$ has a unique element ρ_* of least Hilbert–Schmidt norm: strict convexity gives uniqueness. Every conjugation in $\mathcal{K}$ preserves that set and norm, so ρ_* is invariant. Its observable image belongs to the convex hull of the finite orbit of b and is invariant. Averaging any element of

that orbit over G gives $\bar{b}$; applying this averaging projection to an invariant element in its convex hull shows that its observable image is exactly $\bar{b}$. Therefore the physical channels

$$\widetilde{\Phi}_a(X) = \alpha_a U_a X U_a^* + (1-\alpha_a)\,\mathrm{tr}(X)\rho_*$$

produce (11.3) under every command word. In general ρ_* need not be maximally mixed. The theorem concerns damping toward this common invariant background, not every depolarizing or noisy channel.

11.3. **Periodic plateaux and an explicit rational certificate.** For the periodic mixture of Section S3, let the distinct nonreal frequencies have positive amplitudes γ_j, total $\Gamma = \sum_j \gamma_j$, and let M_* be its exact partition-and-anchor minimum. Its dimension is $d = 2r+1$ and $C_b \le \Gamma/2$. Hence a common damping $\alpha = 1-\epsilon$ retains exactly M_* labels whenever

$$0 \le \epsilon < \min\left\{1, \frac{2\delta_{M_*-1}(b)}{\Gamma(2r+M_*-1)}\right\}. \tag{11.5}$$

This is an effective finite certificate for each algebraic periodic input. It closes the previously existential plateau cutoff for this class, without claiming a sharp transition point or a fixed-memory risk asymptotic.

For clarity, one certificate can be calculated without a general quantifier-elimination run. Consider the unary response

$$b_\alpha(n) = \tfrac{1}{2}\{1 + \alpha^n \cos(\pi n/2)\}, \qquad n \ge 0. \tag{11.6}$$

Proposition 11.3 (A certified four-label interval)**.** *For the undamped sequence $b = b_1$ and the dimension $d = 3$, $\delta_3(b) > 1/36$. In particular the exact stochastic memory of* (11.6) *is four for $89/90 \le \alpha \le 1$. This interval is a certified sufficient interval, not a claimed sharp one.*

Proof. Let Q be any row-stochastic 3×3 matrix, put $t = \mathrm{tr}\,Q$ and $h = \det Q$, and write $y_n = e_1^{\mathsf{T}} Q^n D$ for any scalar decoder. Since $Q\mathbf{1} = \mathbf{1}$, its characteristic polynomial is

$$z^3 - tz^2 + (t+h-1)z - h.$$

Thus $y_{n+3} - t y_{n+2} + (t+h-1)y_{n+1} - h y_n = 0$. We have $0 \le t \le 3$, $|h| \le 1$, and $|t+h-1| \le 3$. The determinant bound follows directly by bounding its absolute permutation sum by the product of the row sums. If the first five values of y are within η of $(1, 1/2, 0, 1/2, 1)$, substitution at $n = 0, 1$ yields

$$|t-h| \le 16\eta, \qquad |2-t-h| \le 16\eta, \qquad |t-1|, |h-1| \le 16\eta.$$

Nonnegativity and the row sums imply a further exact identity. For $a_i = 1 - Q_{ii}$ it is

$$t^2 - 3t + 3 - 3h = \tfrac{1}{2}\sum_{i<j}(a_i - a_j)^2 + 3\sum_{i<j} Q_{ij}Q_{ji} \ge 0. \tag{11.7}$$

To verify it, the second characteristic coefficient is both $t+h-1$ and $\sum_{i<j}(Q_{ii}Q_{jj} - Q_{ij}Q_{ji})$; substitute $Q_{ii} = 1 - a_i$ and simplify. On the other hand, putting $u = t-1$ and $v = h-1$ gives

$$t^2 - 3t + 3 - 3h = -2 - u + u^2 - 3v \le -2 + 64\eta + 256\eta^2.$$

For $\eta \le 1/36$ the right side is at most $-2/81 < 0$, a contradiction. The compact minimum on the first five probes is therefore strictly greater than $1/36$, and so is (11.1), which here uses the six probes $0 \le n \le 5$.

The undamped response has dimension three and a four-cycle realization with decoder $(1, 1/2, 0, 1/2)$. The same polynomial certificate excludes three labels, so its minimum is four. Damping that cycle toward the uniform stationary row realizes (11.6) with four labels. Its difference from the undamped sequence on $0 \le n \le 5$ is at most $\frac{5}{2}(1-\alpha)$. For $\alpha \ge 89/90$ this is at most $1/36$, strictly below $\delta_3(b)$. The finite obstruction proves the lower and completes the proof. □

The certificate uses a stochastic-matrix inequality rather than a sample search over three-label machines. Its accompanying rational diagnostics check the displayed polynomial identity, the constructed all-word damping identities on explicit finite models, and exact finite recurrence consequences. They do not replace the proofs, compute all algebraic gaps, or give an independent human or editorial assessment.

12. Relation to the retained exact resource laws

The finite-group normal form has a cyclic specialization, but it does not replace the independent spectral and arithmetic arguments for responses outside the reversible class. We record the precise relation.

For the probability-valid periodic mixture of Theorem S3.2, with distinct nontrivial frequency orders $q_1, \dots, q_r$, let

$$K = \min_{\mathcal{P}} \sum_{B \in \mathcal{P}} \operatorname{lcm}\{q_j : j \in B\},$$

where $\mathcal{P}$ ranges over partitions of the frequency indices. The exact pure-start count is the minimum of the full period $\operatorname{lcm}(q_1, \dots, q_r)$ and $K + 1$, with the constant and single-frequency endpoints as stated in the retained theorem. A cyclic permutation representation is a disjoint union of cycles; a frequency of order q occurs on a cycle precisely when q divides its length. Assigning required modes to cycles therefore gives the same partition expression. The bounded decoders and positive-amplitude preparation in the retained theorem attain that lower bound. This agreement identifies a model-specific closed form inside the general factorization problem. Arbitrary bounded decoders do not have this same closed formula, as both the sharp S_3 and the five-character contrast transitions show. The quotient theorem applies to arbitrary finite abelian tables; it does not assume the positive-amplitude mixture of the periodic closed form.

The leakage theorem, Theorem S3.5, adds its separately proved inactive live label. Its lower bound uses a decay-normalized observable core and the no-critical-chain argument; the common-background plateau of Section 11 is a different perturbation, as is terminal readout contrast. The exact four-period damping threshold was already $1/\sqrt{3}$ by Theorem S5.3; the current supplement makes this inherited closure explicit. The exact spectral construction of Theorem S5.1, the relative-spectrum obstruction and the arbitrary-phase envelope of Theorem S4.1 remain proved with their own hypotheses. In particular a strictly damped irrational mode need not have compactly reversible observable dynamics.

The following actual-history transfer theorem remains a general statement about causal experiment comparison. It compares complete procedures, charges the common paid acquisition construction, and forms its lower witness on actual competing histories. None of its hypotheses is inferred from a finite observable group. The complete paired supplement retains the associated performed-law minimax and dual theory, finite-width transfer, reset control, optimal discrimination, predictive quotients and geometric results. The original ordered-measurement and two-stage-reset realizations remain independently identifiable by source commit and path.

13. The causal task-order transfer theorem

Suppose that every terminating state-stopped competitor reaches at least $B_0 \geq 1$ live labels under θ_*. Such a certificate must concern actual physical command compatibility, not printed action names. Define

i	$m_-(M)$	$m_+(M)$
cp	M	$\lfloor M/D_0 \rfloor$
ev	M	$\lfloor M/D_0 \rfloor$
st	$M - B_0$	$\lfloor (M-B)/D_0 \rfloor$.

(13.1)

The event upper also requires $M \geq B$. The trivial state-stopped lower $B_0 = 1$ is valid for a procedure initially waiting to make a call.

Theorem 13.1 (Affine task-order transfer with a paid reference and common construction)**.** *Assume all the prepared-kernel, full-report, task-order, effort, occupation, resource and common-construction hypotheses of Section 2. Fix N, M, put $b_- = N/\kappa_-$, $b_+ = N/\kappa_+$, and suppose $b_- \geq 1$ and $m_-(M) \geq 1$. Set*

$$t_N = (1 - A_N)_+ / u_-.$$

If $t_N > 0$, every one of the three risks satisfies

$$R_i^{\Theta}(N, M) \geq \left[l_- t_N \mathcal{Q}_{m_-(M)}(b_-/t_N) - E_N \right]_+. \tag{13.2}$$

If $t_N = 0$, the asserted lower is zero, with no evaluation of $\mathcal{Q}$ at infinity. An empty legal class still has risk $+\infty$.

Suppose also $m_+(M) \geq 1$, $b_+ \geq 1$, and, for event stopping, $M \geq B$. If

$$\beta_N = \frac{l_+ b_+}{1 + b_+ a_+} \geq 1, \tag{13.3}$$

one common Borel program with actual expected cost at most N in every model attains

$$R_i^{\Theta}(N, M) \leq \frac{u_+(1 + b_+ a_+)}{l_+} \mathcal{Q}_{m_+(M)}(\beta_N) + b_+ e_+. \tag{13.4}$$

It commits almost surely. If $\beta_N < 1$, the always-accept first-attempt construction instead gives $R_i^{\Theta}(N, M) \leq u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$ at cost at most $\kappa_+ \leq N$.

The lower is against every legal competing strategy and uses one actual witness. The upper is one parameter-independent strategy for all models. The two effort constants need not agree. Neither direction assumes that the comparison experiment belongs to Θ or that a normalized predictive update is uniformly contracting.

Corollary 13.2 (Scale-uniform relative transfer)**.** *Under Theorem 13.1, suppose $A_N = E_N = a_+ = e_+ = 0$. If $l_+ b_+ \geq 1$, then*

$$\frac{l_-}{u_-} \mathcal{Q}_{m_-(M)}(u_- b_-) \leq R_i^{\Theta}(N, M) \leq \frac{u_+}{l_+} \mathcal{Q}_{m_+(M)}(l_+ b_+). \tag{13.5}$$

The factors $l_\pm, u_\pm$ may be fixed nonunit constants, independent of budget and resolution. In particular no term proportional to $b_\pm$ times a fixed absolute calibration error is required in this relative-order class.

Proof. Substitute $t_N = 1/u_-$ and $\beta_N = l_+ b_+$ in the theorem. □

Corollary 13.3 (Separate exact checkpoint and autonomous laws)**.** *Suppose the certificates are exact: $l_\pm = u_\pm = 1$, all defects vanish, $\kappa_- = \kappa_+ = \kappa$, and exports are the identity. Then, for $N \geq \kappa$,*

$$R_{\mathrm{cp}}^{\Theta}(N, m) = \mathcal{Q}_m(N/\kappa).$$

The constructed autonomous upper uses $B + m$ state-stopped labels, or $\max(B, m)$ event-stopped labels. If the mandatory live count is sharp, $B_0 = B$, then

$$R_{\mathrm{st}}^{\Theta}(N, B + m) = \mathcal{Q}_m(N/\kappa) \quad (m \geq 1), \tag{13.6}$$

$$R_{\mathrm{ev}}^{\Theta}(N, M) = \mathcal{Q}_M(N/\kappa) \quad (M \geq B). \tag{13.7}$$

The subtraction by B belongs only to the state-stopped model. Nonidentity exports require the expanded dictionary in (13.1)*, unless an optimal dictionary is invariant under all exports. These exact stopping distinctions are inherited from the complete companion and are reproved here to fix the resource semantics.*

Proof. The same lower and upper profile coincide in each asserted interface. Identity exports give $D_0 = 1$. The packing and converse widths are proved in Lemma 14.4. No lower on live states is used to forbid event-stopped reuse. □

The familiar absolute-defect theorem is also a specialization: take $l_\pm = u_\pm = 1$ while retaining A_N, E_N, a_+, e_+. Thus relative and absolute calibration are composed in one statement. The chosen metric is the common scored task throughout; it is not changed between the converse and the construction.

14. Proof of the central theorem

Lemma 14.1 (Profile attainment)**.** *For every finite $m \geq 1$ and $b \geq 1$, the minimum in* (2.2) *is attained. The profile is nonincreasing in m, b and left-continuous in $b > 1$.*

Proof. Write a dictionary as an ordered tuple in the compact set H^m, permitting repetitions. The feasible densities are a weak-star closed subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are each closed under pairing with $L^1(\nu)$. They are therefore weak-star compact. As the tuple varies, its squared-distance function varies uniformly on H, since all points belong to a bounded set. Along a convergent product net, split the objective difference into this uniform variation and the weak-star pairing against the fixed bounded measurable loss at the limit tuple. This proves continuity of the objective and attainment. Nets avoid an unwarranted sequential compactness assumption. Monotonicity follows from inclusion of feasible sets. For left continuity take an optimizer h at $b > 1$ and let $h_\lambda = (1-\lambda)h + \lambda$. It has unit mass and cap $b_\lambda = (1-\lambda)b + \lambda < b$. Its objective tends to the optimum as $\lambda \downarrow 0$, while monotonicity bounds the optimum at b_λ between the optimum at b and this objective. □

Lemma 14.2 (Actual-history comparison measure)**.** *Fix any robustly admissible competitor of cost at most N, with its fixed terminal dictionary C. There are finite nonnegative measures ξ_c on X, $c \in C$, such that $\xi = \sum_c \xi_c$ obeys*

$$\xi \leq (\mathbb{E}_* A_\tau)\nu \leq b_- \nu, \tag{14.1}$$

$$1 \leq u_- \xi(X) + A_N, \tag{14.2}$$

$$\mathbb{E}_* \|Z - \widehat{r}\|^2 \geq l_- \sum_{c \in C} \int d_c \, d\xi_c - E_N. \tag{14.3}$$

The measure ξ is not asserted to be the terminal law of any comparison execution.

Proof. Adjoin an independent uniform random variable to the current report on both sides. This is current-call randomness, not a seed retained between calls; it is sufficient to represent the competitor's randomized output and stopping decision. A finite-output kernel has such a representation by cumulative probabilities. At a fixed reached productive history h and chosen command a, the post-report decision then gives an acceptance indicator $q_h(y)$ and an output $r_h(y) \in C$ on acceptance. Give r_h an arbitrary fixed value on rejection. The notation y now includes the current independent random variable. Integrating the original certificates over that variable preserves their constants. Dependence on the entire earlier history is permitted in these functions.

For each Borel $D \subset X$, set

$$\xi_c(D) = \mathbb{E}_* \sum_{j \leq \tau:\ j \ mproductive} \int \mathbf{1}_{\{\pi_{H_{j-1}, a_j}(y) \in D\}} q_{H_{j-1}}(y) \mathbf{1}_{\{r_{H_{j-1}}(y) = c\}} P^0_{H_{j-1}, a_j}(dy, H). \tag{14.4}$$

The outer expectation is the actual pre-call occupation at θ_*. All terms are jointly measurable and nonnegative. Tonelli defines the measures, whose total is finite because $\mathbb{E}_* A_\tau \leq b_-$. Summing over c and dropping the acceptance indicator gives (14.1): each comparison row has mark law ν.

The analogous sum with the actual rows instead of P^0 has mass exactly one, because the actual procedure commits exactly once almost surely. Apply (2.4) at each actual reached history, then

sum. The occupation certificate yields (14.2). Apply (2.5) in the same way to the disjoint actual commitment events. Their expected losses sum to the actual risk. The comparison contribution is $d_{r_h(y)}(\pi_{h,a}y)$ by (2.3); this identity conditions on the full current report and therefore does not overlook extra information seen by the competitor. The accumulated loss defect is at most E_N, giving (14.3). One may first truncate the number of calls. Monotone convergence for the nonnegative contributions and the finite defect bounds justify the limit.

No actual past is replaced by a comparison past. In particular the competitor need not terminate, or obey a cost bound, if hypothetically run under the comparison experiment. This distinction is why the argument applies when that experiment is not an operating model. □

Remark 14.3 (Why the full report is required). A constant mark can conceal a visible fair bit which also equals the future audit. A comparison using only that mark has $f = 0, w = 1$, but a two-value decoder reading the bit predicts it with zero loss. The hypotheses therefore condition on the full report seen by the competitor; matching a compressed mark marginal alone cannot establish the lower.

Lemma 14.4 (Live roles, output roles and physical compatibility)**.** *A program with ℓ live roles and t output roles has a state-stopped implementation with $\ell+t$ labels and an event-stopped implementation with $\max(\ell, t)$ labels, preserving every path cost and complete stopped scored law. Conversely an event-stopped M-label program has at most M decoder values. A state-stopped M-label program with at least B_0 reached live labels has at most $M - B_0$ terminal decoder values.*

For a physical live-state certificate, consider finitely many mandatory pre-call contexts I with no free context input. Suppose the commands legal at a history of context i are exactly $\mathcal{U}_i$, and every terminating competitor visits every context with positive probability under the same witness. This positive probability may depend on the competitor; no uniform lower bound over competitors is required. Let b_{phys} be the least size of a cover of I by subsets J with $\bigcap_{i\in J}\mathcal{U}_i \neq \varnothing$. Then at least b_{phys} live labels are required. This certificate is invariant under duplicate names and bijective stateless recodings of the same physical commands.

Proof. For state stopping take disjoint copies of the live and terminal role sets and type them zero and one. For event stopping embed the live and terminal role sets separately into one alphabet of size $\max(\ell, t)$. On continuing transitions use the live embedding; on commitment emit the final event and use the terminal embedding. A command row is read only on a continuing branch, while the decoder is read only after commitment. Sharing their label therefore causes no conflict and uses no continuing mode bit. Couple the original and recoded coins. Every finite stopped prefix and every path cost agree, and almost-sure termination gives equality of the stopped scored laws. A fixed decoder has one value per retained label. In state stopping the global halt map excludes each reached live label from the output set, proving its subtraction. Event stopping has no such exclusion.

A live label reached in contexts J must have one command distribution supported on all the corresponding $\mathcal{U}_i$. Since J is finite, the union of the exceptional null command sets is null; hence their intersection supports a probability and is nonempty. The sets of contexts served by the labels cover I, proving the bound. Conversely a nonempty intersection contains a command and hence a Dirac command distribution. Replacing names without changing executed kernels, legal domains, costs and reports does not change these intersections. A translator which itself remembers a hidden phase is not such a recoding; its memory must be charged. □

Proof of Theorem 13.1. Fix a competing program. Lemma 14.4 bounds its decoder width by $m_-(M)$. Let $s = \xi(X)$ in Lemma 14.2. If $t_N > 0$, then (14.2) gives $s \geq t_N$. Scale the measure to $\xi' = (t_N/s)\xi$. It has mass t_N and remains bounded above by $b_-\nu$. Thus $h = (d\xi'/d\nu)/t_N$ has unit mass and cap

b_-/t_N. Nearest-dictionary projection and nonnegativity imply

$$\sum_{c\in C}\int d_c\,d\xi_c \geq \int [w+\operatorname{dist}(f,C)^2]\,d\xi \geq t_N\mathcal{Q}_{m_-(M)}(b_-/t_N).$$

Together with (14.3) this proves the asserted lower for this competitor. Its worst-model risk dominates its risk at θ_*. Taking the infimum over competitors gives (13.2), without exchanging an infimum and a supremum. Nonnegative loss gives the zero truncation and the stated case $t_N = 0$.

For the upper assume (13.3). Choose an attained optimizer (C,h) at width $m_+(M)$ and cap β_N. Let $r_C(x)$ be the first nearest point to $f(x)$ in a fixed ordering of C. After each attempt of $\mathcal{G}$, accept x with probability $h(x)/\beta_N$, export $T_J r_C(x)$ and commit; otherwise return to the start routing label. All these operations are Borel. The choice J is used within the current attempt; any part retained thereafter is charged through the live role or exported decoder, not a free seed. The exports are assumed jointly measurable and take at most $D_0|C|$ distinct values, so their finite decoder and transitions are Borel as well.

Conditional on every surviving attempt history, the acceptance probability is at least

$$p = \frac{l_+}{\beta_N} - a_+ = \frac{1}{b_+} > 0. \tag{14.5}$$

Consequently the probability of reaching attempt $j+1$ is at most $(1-p)^j$, by induction on conditional survival probabilities. In particular commitment is almost sure and the expected number of attempts is at most $1/p = b_+$. Each begun attempt has conditional expected cost at most κ_+, not merely a cost conditional on a favorable outcome. Summing the predictable attempt costs gives actual expected cost at most $\kappa_+ b_+ = N$.

The conditional unnormalized loss contributed by an attempt is at most $u_+\mathcal{Q}_{m_+(M)}(\beta_N)/\beta_N + e_+$. Sum its contributions on the disjoint terminal events, using the same survival bound. The result is at most

$$b_+\left[\frac{u_+}{\beta_N}\mathcal{Q}_{m_+(M)}(\beta_N) + e_+\right],$$

which is (13.4). The program has B live roles and at most $D_0 m_+(M)$ terminal roles. Lemma 14.4 gives exactly the sufficient widths in (13.1). A checkpoint counts only the terminal width. No physical renewal law was used: the conditional hazard controls survival even if the hidden state after rejection differs arbitrarily from its former state.

If $\beta_N < 1$, commit after the first attempt, and use an optimizer at cap one. The loss certificate with $q = 1$ gives $u_+\mathcal{Q}_{m_+(M)}(1) + e_+$, and the whole-attempt cost is at most κ_+. This proves all remaining assertions. □

15. Scope and further mathematical questions

The compactly reversible normal form is a classification of exact input–response realization. The contrast theorem applies to every finite observable group, while the quotient linear programs apply to groups whose subgroups are normal. The sharp S_3 and $(\mathbb{F}_2)^4$ thresholds are complete solutions of specified models. They do not identify every nonabelian positive-factorization minimum. The full contrast filtration is finite and algebraically computable for algebraic data; a closed formula for all groups or a polynomial synthesis algorithm remains a different question.

The retained posterior-positivity theorem handles arbitrary finite collections of known instruments satisfying its uniform hypothesis, including state-dependent observations and feedback from visible histories. It gives one exact simulator for a fixed known initial state. It does not convert an unknown quantum input into a classical label, learn unknown primitives, or produce parameter-independent tables for an arbitrary statistical model. The retained geometric lower bound is for a physical dictionary covering a whole interior state body. Its matching small-noise exponent is not a lower

bound for the unrestricted memory of each individual instrument. The rank-three family instead has exact unrestricted memory lower bounds, including the separate charge for pure initialization.

Terminal readout contrast, channel damping repeated after each command, and noise on every observed instrument branch are different operations. The four-period damping transition at $1/\sqrt{3}$ was already determined by the retained first-sector theorem; Corollary S1.1 records that consequence explicitly. The previous finite-probe gap remains valid. Other damping transitions, intermediate phase-type timing frontiers and the general M squared-sign risk curve are not settled here. The exact periodic partition and leakage laws keep their spectral and probability hypotheses. The arbitrary-phase arithmetic envelope keeps both logarithmic exponents and its badly-approximable criterion; it supplies no particular irrationality exponent for the algebraic phase $(3+4i)/5$.

Full-history optimization, finite-memory attainment, finite-memory density, approximation at a fixed memory budget and exact response simulation are distinct problems. The robust tied criterion already proved on the rigid herald class remains in force. General nonrigid boundary ties and a classification of optimal several-command feedback remain open. Exact algebraic descriptions of transition tables also leave sampling, precision, program length and working storage as additional resources.

The actual-history transfer theorem and the performed-law, variational, dual, spectral, reset, discrimination and geometric results retain complete proofs. The independent absolute-calibration companion and the original finite-order measurement and two-stage reset realizations retain their source identities. Claims requiring A1, A2 or further Theta construction hypotheses remain independent research objectives. The scientific question throughout is how the information acquired by a causal experiment constrains the resources required to realize and use its predictive state.

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